Learning a Self-Similar Grammar from One Example: Tied Random Hierarchy Models, Context Clustering, and the Value of Scale Symmetry

Companion note to "Self-Similar Recursive Grids" (srg.pdf), Open Problem 9.2

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Abstract

The Random Hierarchy Model (RHM) is a synthetic compositional task: a context-free grammar of depth L with v symbols, m synonymous rules per symbol, and branching s. Backprop-trained deep CNNs need $P^* \propto v\, m^L$ labelled examples [1]. Latent-prediction methods need $\asymp vm^3$ examples, which is independent of L [3]. In both lines of work the rules are drawn *independently at every level*. We study the *tied* (self-similar) RHM, in which one grammar is used at every level. This is the stochastic-process counterpart of the scale-tied recursive grid. Three results follow. **(1) Without tying, labels are unavoidable.** In the untied RHM, *every* learner has root accuracy at most $1/v + P/(vm)$ from P labelled examples. This holds even with unlimited unlabelled data and an oracle for all lower-level rules, because the top rule has no context. **(2) With tying, clustering bottom tuples by their sibling context recovers the grammar.** This rule is a statistical form of the Myhill–Nerode quotient. Given the latents at height ≥ 2, the s^{L-2} bottom grids of each example are independent, so correlations inside a tree enter only through one non-degeneracy condition. The number of examples needed falls like $s^{-(L-2)}$. **(3) The clusters are named by self-similarity.** The consistent namings are exactly a coset of the automorphism group of the valid-tuple set, which is trivial with high probability. Together: from a *single unlabelled example* of sufficient depth, the learner recovers the grammar exactly and classifies the root perfectly, without labels or backpropagation. Experiments (v=8, m=4, s=2) confirm this. Exact recovery needs about 1024–2048 bottom grids however they are packaged into trees, and one example suffices from $L = 11$ on. Backprop through untied or tied CNNs with root labels moves the other way: its label requirement roughly doubles per level (512 at $L = 2$, 16384 at $L = 6$), and the tied architecture alone does not help it. The gap between backprop's labelled examples and the context learner's unlabelled examples is therefore $1\times$ at $L = 2$ and $512\times$ at $L = 6$. **(4) Partial tying degrades gracefully.** If each rule at each level is reused with probability ρ and replaced otherwise, every zero-label learner has error at least $\approx (1-\rho)(1-1/v)$, the root's own unexplained rules. A level-wise learner attains this floor up to an $O((1-\rho)vm\log(vmL)/P)$ term, and *neither bound depends on depth*: the number of nodes at a level controls both how easily its deviations are learned and how often they matter, and the two effects cancel. Experiments reach the floor at every depth for $0.6 \leq \rho \leq 0.9$, and the breakdown near $\rho = 0.5$ is explained. **(5) The gains are not an artifact of discreteness.** In a linear-Gaussian self-similar tree, the same pipeline carries over: sibling moments whose within-tree correlations cancel, and naming as fixing the latent basis by self-similarity (unique up to the Gaussian sign symmetry). It learns from one unlabelled tree of depth 9, while an untied tree provably needs labels. What does *not* carry over is backprop's exponential label cost, which is a combinatorial effect: in the linear model backprop needs a few hundred labels at any depth tested. **(6) The mechanism is general.** For an arbitrary tied tree process (noisy, continuous, nonlinear), context identifies each grid's belief about its parent, i.e. its belief-propagation message; estimating it depends on the data only through the

total number of bottom grids; and naming is identified up to the automorphisms of the kernel, which additive Gaussian noise reduces to a finite-dimensional group of isometries commuting with the sibling maps. For bounded nonlinear Gaussian trees, the laws of the bottom two levels (hence one deep tree) determine the noise, the kernel and its naming up to that group, with no labels (§8, Theorem 36). **(7) Why backprop slows with depth.** The correlation between the root label and any local feature of the leaves decays like $q_1^{L/2}$, where $q_1 \le s\lambda^2$ is a linear Kesten–Stigum number of the kernel. The first phase of gradient descent sees only this signal. Gaussian trees have no gap (their Bayes predictor is linear); tied grammars with $q_1 < 1$ do. For the grammars above $q_1 \approx 0.4$, and $1/q_1$ matches the measured growth of backprop's labels per level (§9). **(8) Real images.** On CIFAR-10 the state of a region is its 2×2 thumbnail, which is parent-determined and observed, together with a hidden context type that context theory requires (siblings stay dependent given the thumbnail). Sum–product over types is then exact, and the tree is an exact density on images. With zero labels it reaches 4.75 bits/dim, and its samples reproduce the per-scale heavy tails of natural images. Its excess loss from N unlabelled images is $\approx 2.5k/(2N)$ spread over all scales, while an untied top pays the predicted $492/N$ bits per coordinate. About 10^3 images give the kernel that $5 \cdot 10^4$ give. A linear head on the root is the Bayes readout, but a likelihood-trained root holds little class information (0.30–0.38). The top scale is where CIFAR stops being self-similar (§10). The class information is not lost in the tree but at its last merge: the four quadrant states give 0.55, and the hidden types add to the thumbnails at every scale. Observing labels at the root in the joint likelihood costs at most $H(Y)$ per image in density, and it can therefore move only what the pixels leave undecided. The root type is already decided by the pixels, so labels add little ($0.17 \to 0.25$ with an untied top). An observed label's influence on a node falls about twentyfold per level on CIFAR, and it bounds the root-label backprop gradient there as well (§10.8).

1 Models

Let $V = \{1, \dots, v\}$ and fix a branching $s \ge 2$. A *grammar* $G = (R_a)_{a \in V}$ assigns to each symbol m rules $R_a \subset V^s$, $|R_a| = m$, and the sets are pairwise disjoint (unambiguity). Write $D = \bigcup_a R_a$ ($|D| = vm$) for the valid tuples, $f = m/v^{s-1} \in (0, 1]$ for their fraction, and $g : D \to V$ for the parent map. For the random grammar, D is a uniformly random vm-subset of V^s, partitioned uniformly into v blocks of size m.

Definition 1 (tied and untied RHM)**.** Draw the root $z_\epsilon \sim \mathrm{Unif}(V)$. Every internal node w independently picks a rule uniformly from $R_{z_w}^{(\mathrm{ht}(w))}$, and its s children take the symbols of that rule. The input is the s^L leaf symbols and the label is $y = z_\epsilon$. In the *untied* RHM of [1, 3], $R^{(h)}$ are independent random grammars. In the *tied* RHM (SRHM), $R^{(h)} = R$ at every height.

For both models, knowing the grammar(s) makes classification trivial: parse bottom-up with g. For the tied model this is a discrete scale-tied recursive grid with state space $V \cup \{\perp\}$ and one table g (srg.pdf, §3). By the Myhill–Nerode theorem for grids, the minimal grid for the task has exactly the grammar's symbols as states. *Learning the grammar means finding the synonym partition of D and naming its blocks.*

2 Without tying, labels are unavoidable

Theorem 2. *In the untied RHM with a uniformly random top grammar $R^{(L)}$, consider any learner that receives P labelled examples, any number of unlabelled examples, and the lower-level grammars $R^{(1)}, \dots, R^{(L-1)}$ exactly. Its expected root accuracy on a fresh example is at most*

$$\frac{1}{v} + \frac{P}{vm}.$$

Accuracy $1-\varepsilon$ therefore needs $P \geq (1-\frac{1}{v}-\varepsilon)\,vm$ labels.

Proof. Given the lower grammars, the leaves determine the top tuple $u \in D_L$ (the root's children), and nothing more is relevant. For every example, u is uniform on D_L. Conditioned on u and the root, the rest of the tree does not depend on how D_L is partitioned. Unlabelled data therefore reveal at most D_L. By exchangeability of the random partition, the posterior given D_L and the $U \leq P$ distinct labelled top tuples is uniform over the partitions consistent with them. A test tuple u^* is one of the labelled tuples with probability $U/(vm)$. Otherwise its parent is a with probability $(m-k_a)/(vm-U) \leq m/(vm-U)$, where k_a counts the labelled tuples with parent a. The success probability is at most $\frac{U}{vm} + \big(1-\frac{U}{vm}\big)\frac{m}{vm-U} = \frac{U}{vm} + \frac{1}{v}$. □

The top grid is the one grid without context. Nothing below it can reveal which tuples are synonyms at the top, and only labels can. Latent-prediction learners [3] recover levels $1,\dots,L-1$ without labels, but the last level still costs $\Theta(vm)$ labels. *Tying removes this barrier.* The top level uses the same partition as the bottom level, where every tuple has abundant context.

3 With tying: context clustering at the bottom

Let $\mathcal{H}_2$ be the latents at all heights ≥ 2, and let b_w be the symbol of a height-2 node w (a *grid* whose s children are bottom tuples). Let n_b be the number of grids with $b_w = b$, pooled over the P examples, and $n_2 = \sum_b n_b = Ps^{L-2}$. For $b \in V$ and ordered positions $i \neq j$, let $c_b(a,a') = \#\{r \in R_b : (r_i, r_j) = (a,a')\}$ and $c_b(a) = \sum_{a'} c_b(a,a')$. Given a histogram $n = (n_b)$, the *realized sibling law* is

$$\nu^n(a' \mid a) = \frac{\sum_b n_b\, c_b(a,a')}{\sum_b n_b\, c_b(a)}, \qquad \Delta(n) = \min_{a_1 \neq a_2} \|\nu^n(\cdot \mid a_1) - \nu^n(\cdot \mid a_2)\|_1.$$

For a bottom tuple t, let $\hat{\mu}_t$ be the empirical distribution of the tuple at position j over all grids having t at position i.

Lemma 3 (statistical Myhill–Nerode)**.** *Conditioned on $\mathcal{H}_2$, the grids are independent. For $t \in R_a$, the conditional mean of $\hat{\mu}_t$, reweighted by selection probabilities, is*

$$\mu_t^n = \sum_{a'} \nu^n(a' \mid a)\ \mathrm{Unif}(R_{a'}),$$

which depends on t only through its parent a. *Synonymous tuples have identical context laws, and tuples with parents $a_1 \neq a_2$ have context laws at L^1 distance $\|\nu^n(\cdot|a_1) - \nu^n(\cdot|a_2)\|_1 \geq \Delta(n)$.*

Proof. Given $\mathcal{H}_2$, grid w picks $r \sim \mathrm{Unif}(R_{b_w})$, and its children pick $t_{wk} \sim \mathrm{Unif}(R_{r_k})$ independently. So $P(t_{wi} = t, t_{wj} = t') = c_{b_w}(a,a')/m^3$ for $t \in R_a$, $t' \in R_{a'}$. Summing over grids and normalizing gives μ_t^n. The L^1 identity holds because the $\mathrm{Unif}(R_{a'})$ have disjoint supports. □

This is the context-equivalence of srg.pdf, Thm. 3.2, in statistical form. Two grids are merged when *no context distinguishes them*, and here the context is the sibling. It is also the "copy grids with equal histories" rule of the original notes.

Theorem 4 (partition recovery; correlation-free)**.** *Condition on $\mathcal{H}_2$ with $\Delta(n) \geq \Delta_0 > 0$. Let $\gamma_a = \sum_b n_b c_b(a)/(m\,n_2)$ be the realized frequency of parent a at position i, and $\gamma_{\min} = \min_a \gamma_a$. There is an absolute constant C such that, if*

$$n_2 = P\,s^{L-2} \ \geq\ C\,\frac{v\,m^2 \log(vm/\delta)}{\Delta_0^2\,\gamma_{\min}},$$

then with probability $\geq 1-\delta$ *every* $t \in D$ *satisfies* $\|\hat{\mu}_t - \mu_t^n\|_1 \leq \Delta_0/4$. *Single-linkage clustering of* $\{\hat{\mu}_t\}$ *into* v *clusters then returns exactly the synonym partition of* D.

Proof. Fix $t \in R_a$ and write $X_w = \mathbf{1}[t_{wi} = t]$ and $p_w = \mathbb{E}X_w = c_{b_w}(a)/m^2$. Then $\mathbb{E}N_t = \sum_w p_w = n_2\gamma_a/m$. Put $A = \sum_w (X_w \mathbf{e}_{t_{wj}} - p_w \ell_w)$, with ℓ_w the conditional law of t_{wj} given $X_w = 1$, and $B = N_t - \mathbb{E}N_t$. Both are sums of independent, centred, bounded terms with variance proxy $\leq \mathbb{E}N_t$, and $\hat{\mu}_t - \mu_t^n = (A - B\mu_t^n)/N_t$. A vector Bernstein inequality [4] and a Chernoff bound ($N_t \geq \mathbb{E}N_t/2$) give $\|\hat{\mu}_t - \mu_t^n\|_2 \leq C'\sqrt{\log(1/\delta')/\mathbb{E}N_t}$ with probability $1-\delta'$. Since $\|\cdot\|_1 \leq \sqrt{vm}\,\|\cdot\|_2$, the condition $\mathbb{E}N_t \geq C'' vm \log(1/\delta')/\Delta_0^2$ suffices. Take a union bound over the vm tuples with $\delta' = \delta/(vm)$. Synonyms then lie within $\Delta_0/2$ of each other and non-synonyms at least $\Delta_0/2$ apart (by Lemma 3). Each block has m members at mutual distance $< \Delta_0/2$, so single-linkage merges every block before any two blocks. □

Corollary 5 (examples needed fall exponentially with depth)**.** *If* $\Delta(n) \geq \Delta_0$ *and* $\gamma_{\min} \geq \gamma_0$ *on the realized* $\mathcal{H}_2$*, then*

$$P^*_{\text{tied}}(L) \;\leq\; \left\lceil C\,\frac{v\,m^2\log(vm/\delta)}{\Delta_0^2\,\gamma_0}\; s^{-(L-2)} \right\rceil,$$

so a single example *suffices once* $L \geq 2 + \log_s\big(Cvm^2\log(vm/\delta)/(\Delta_0^2\gamma_0)\big)$.

Correlations inside a tree enter *only* through the realized histogram n, by way of $\Delta(n)$ and $\gamma_{\min}$. This mirrors the kernel-learning result for broadcast processes (srg.pdf, Thm. 5.6). Both are continuous functions of n/n_2 away from degenerate histograms, so they are close to their population values whenever the histogram concentrates. The following proposition makes that quantitative.

Proposition 6 (histogram concentration; non-CI extension of srg.pdf Thm. 5.5)**.** *Let* λ *be the largest maximal correlation of a parent–child pair and* λ_{sib} *the largest maximal correlation of a sibling pair (*$i \neq j$ *children of one node). Let* $q = s\lambda^2$. *For any function* ϕ *of a height-2 node's latent with* $\operatorname{Var}\phi \leq V$,

$$\operatorname{Var}\Big(\tfrac{1}{s^{L-2}} \sum_{\text{ht}(w)=2} \phi(b_w)\Big) \leq \frac{V}{s^{L-2}}\Big[1 + (s-1)\lambda_{\text{sib}} \sum_{j=0}^{L-3} q^j\Big].$$

Proof. For $u \neq w$ at height 2 with lowest common ancestor at height $2+j$, the path $\phi(b_u) - z_u - \cdots - z_{c_u} - z_{c_w} - \cdots - z_w - \phi(b_w)$ is a Markov chain. Here c_u, c_w are the children of the LCA, whose joint law is a sibling link. Its maximal correlation is at most $\lambda^{2(j-1)}\lambda_{\text{sib}}$ by submultiplicativity. There are $(s-1)s^{j-1}$ such w per u. □

In the conditionally independent (broadcast) case $\lambda_{\text{sib}} \leq \lambda^2$, which recovers srg.pdf, Thm. 5.5. Combined with continuity of $\Delta(\cdot)$ and $\gamma(\cdot)$, the proposition bounds $P(\Delta(n) < \Delta_0)$ by Chebyshev. This is a sufficient condition with a worst-case λ, and it is conservative: in the experiments the realized histogram of a *single* deep tree was always non-degenerate.

4 Naming by self-similarity

Clustering yields the blocks with arbitrary ids $k \in \{1, \ldots, v\}$. Parsing the level above requires the map $\pi : \text{ids} \to V$, because the level-1 latents are *the same kind of symbol* as the leaves. For each height-2 grid, map its children's tuples to ids. The resulting id-tuples are π_0^{-1} of the true level-1 tuples, which are valid.

Theorem 7 (naming)**.** *Suppose the partition is correct and every $d \in D$ occurs as the true children tuple of some height-2 grid. Then π maps every observed id-tuple into D iff $\pi\pi_0^{-1} \in \mathrm{Aut}(D) := \{\tau \in S_V : \tau^{\otimes s}(D) = D\}$. In particular, the naming is identified with* zero labels *iff $\mathrm{Aut}(D)$ is trivial. If it is not trivial, at most $|\mathrm{Aut}(D)|$ candidates remain for labels to decide between.*

Proof. The observed id-tuples are exactly $\pi_0^{-1}(D)$. So π is consistent iff $\tau(D) \subseteq D$ with $\tau = \pi\pi_0^{-1}$, which by cardinality means $\tau(D) = D$. □

Lemma 8 (random valid sets are asymmetric)**.** *Let D be a uniformly random subset of V^s of density $f \in (0,1)$, and put $\beta = f^2 + (1-f)^2 < 1$. Then*

$$P\big(\mathrm{Aut}(D) \neq \{\mathrm{id}\}\big) \;\le\; C\, v^{s/2} \cdot 2\big(v\,\beta^{v^{s-1}/2}\big)^2 \qquad \textit{whenever } v\beta^{v^{s-1}/2} \le \tfrac{1}{2}.$$

Proof. First work in the Bernoulli model, where each tuple is valid independently with probability f. A permutation τ moving k symbols moves $u \ge v^s - (v-k)^s \ge kv^{s-1}$ tuples. $\tau(D) = D$ requires every $\tau^{\otimes s}$-cycle of length $\ell \ge 2$ to be all valid or all invalid, which has probability $f^\ell + (1-f)^\ell \le \beta^{\ell/2}$ (the ℓ-norm is at most the 2-norm). Cycles are disjoint, so $P(\tau \in \mathrm{Aut}) \le \beta^{u/2} \le \beta^{kv^{s-1}/2}$. A union over the at most v^k permutations moving k symbols, summed over $k \ge 2$, gives $2(v\beta^{v^{s-1}/2})^2$. Passing to the fixed-size model costs at most a factor $1/P(\mathrm{Bin}(v^s, f) = vm) \le Cv^{s/2}$. □

The bound is asymptotic: it is vacuous at $v = 8$, $s = 2$, and meaningful from about $v = 32$. We therefore also computed $|\mathrm{Aut}(D)|$ exactly for every grammar used; all were trivial (Table 1).

Corollary 9 (one-example, zero-label learning of a self-similar grammar)**.** *In the tied RHM, suppose $\mathrm{Aut}(D)$ is trivial and the realized histograms are non-degenerate ($\Delta_0, \gamma_0 > 0$). Then context clustering followed by naming by self-similarity recovers the grammar exactly, and classifies every root correctly, from $P = O\big(vm^2 \log(vm/\delta)\, \Delta_0^{-2} \gamma_0^{-1} s^{-(L-2)}\big)$ unlabelled examples and zero labels. By contrast, in the untied RHM every learner needs $\Omega(vm)$ labels (Theorem 2).*

5 Experiments

We use $v = 8$, $m = 4$, $s = 2$ ($f = 1/2$, $|D| = 32$) and 8 random grammars. The practical learner refines a spectral (co-occurrence) initialization by hard EM on sibling cluster-ids, uses capacity-m balanced assignment and 15 restarts, and selects among restarts without labels (the naming-consistency count first, then likelihood). Naming is by brute force over S_8. The code is in `verify/srhm.py` and `verify/exp_srhm.py`.

Table 1: Channel constants of the 8 grammars (max over all edges of a depth-8 tree) and automorphism counts.

grammar	λ	λ_{sib}	$q = s\lambda^2$	$\lvert\mathrm{Aut}(D)\rvert$
0	0.801	0.712	1.28	1
1	0.760	0.749	1.16	1
2	0.971	0.667	1.88	1
3	0.904	0.788	1.63	1
4	0.916	0.740	1.68	1
5	0.812	0.751	1.32	1
6	0.687	0.687	0.94	1
7	0.886	0.747	1.57	1

Table 2: Zero-label context learner on the tied RHM: fraction of 8 grammars recovered *exactly* (root accuracy = 1), by depth L and number of unlabelled examples P. Parentheses give the number of bottom grids $P\,s^{L-1}$.

L	$P=1$	$P=2$	$P=4$	$P=8$	$P=16$	$P=32$
5	0 (16)	0 (32)	0 (64)	0 (128)	0 (256)	0 (512)
6	0 (32)	0 (64)	0 (128)	0 (256)	0 (512)	**0.75** (1024)
7	0 (64)	0 (128)	0 (256)	0 (512)	**0.75** (1024)	**1** (2048)
8	0 (128)	0 (256)	0.12 (512)	**0.88** (1024)	**1** (2048)	**1** (4096)
9	0 (256)	0 (512)	**0.88** (1024)	**1** (2048)	**1** (4096)	**1** (8192)
10	0 (512)	**0.88** (1024)	**1** (2048)	**1** (4096)	**1** (8192)	–
11	**0.88** (1024)	**1** (2048)	**1** (4096)	**1** (8192)	–	–

Table 3: Sample complexity on the *same* tied RHM. Backprop: smallest P (powers of 2) of labelled examples with test accuracy ≥ 0.9 (mean of 2 seeds; Adam, width 256), for a standard deep CNN (kernel = stride = s, untied layers) and for the scale-tied CNN (one cell map applied L times). Context learner: smallest P of *unlabelled* examples with exact recovery in $\geq 6/8$ grammars. Full logs are in `verify/exp_cnn*.log`.

	backprop, root labels: P^* (acc ≥ 0.9)		context learner, 0 labels	ratio
L	untied CNN	tied CNN (SRG)	P^* unlabelled (success ≥ 0.75)	labels / unlabelled
2	512	512	512	1
3	2048	2048	256	8
4	4096	4096	128	32
5	8192	8192	128	64
6	16384	> 65536 (not trained; best 0.34)	32	512
7–11	(not run)	(not run)	16, 8, 4, 2, **1**	–

Reading the tables. (i) Exact recovery by the zero-label learner is governed by the *total number of bottom grids*, not by the number of examples. At every depth, success jumps between 1024 and 2048 grids (Table 2; at $L \leq 5$ we tested only the 1024/2048/4096 points, with success 0.62–0.88, then 1, then 1). The unlabelled examples needed therefore halve with every level, and a *single* example of depth 11 (2048 leaves) recovers the grammar in 7/8 grammars, as Corollary 5 predicts. The worst-case bound of Prop. 6 is far more pessimistic, since all but one grammar have $q > 1$; the correlation-free Theorem 4 is the relevant description. Naming was unique in every successful run, consistent with $|\operatorname{Aut}(D)| = 1$. (ii) Backprop from root labels moves the other way: P^* roughly *doubles* per level, from 512 at $L = 2$ to 16384 at $L = 6$. At this grammar size the growth is slower than the m^L reported for the untied RHM at other parameters [1]. Giving backprop the scale-tied architecture does *not* help. The tied CNN matches the untied one up to $L = 5$, and at $L = 6$ it failed to train within our budget, which we attribute to optimization of a depth-6 recurrence without normalization rather than to the statistics. The gap between labelled examples for backprop and unlabelled examples for the context learner grows from $1\times$ at $L = 2$ to $512\times$ at $L = 6$. (iii) Caveats: one grammar size, two seeds per backprop point, a fixed training budget, and no hyperparameter search for the CNNs. The backprop numbers are indicative; the context-learner numbers are the object of the theorems.

6 Partial tying: graceful degradation

Real data are at best approximately self-similar. We therefore interpolate between the two models.

Definition 10 (PT-RHM(ρ))**.** Fix a base grammar R, used at height 1, with valid set D and $f \le 1/2$. At every height $h \ge 2$, each of the vm rule *slots* (a, j) independently keeps its base rule $R_{a,j}$ with probability ρ. Otherwise it receives a *fresh* tuple, drawn without replacement (per height) from $V^s \setminus D$. Let K_h be the number of fresh slots at height h and $k_{h,a}$ the number belonging to class a. $\rho = 1$ is the tied RHM; $\rho = 0$ is an untied RHM whose rules avoid D.

Learner. (1) Learn the base partition at the bottom by context clustering (§3). (2) Name the clusters *structurally* (Prop. 14). (3) For $h = 2, \dots, L-1$, parse the level below. A tuple at height h outside the learned D is a deviation. Once it has been seen N_0 times with a parsable sibling, assign it to the class whose sibling-context profile at height h it matches best. The profiles are estimated from base tuples at the same height with the same filtering, so any selection bias cancels. Assignment is *capacity-constrained*. Every class has exactly m rules per height, so class a has $m -$ (base rules of a observed at height h) fresh slots, a label-free quantity. A few EM rounds then re-estimate the profiles including the assigned deviations. Without the capacities, a class that keeps none of its base rules at some height has no profile, and its deviations are misassigned. At $\rho = 0.6$ this happened, and the errors cascaded upward. With the capacities, all deviations were assigned correctly. (4) At the root there is no context: assign fresh root tuples by elimination, to the class with the most base rules never seen at the top.

Theorem 11 (zero-label floor; any learner)**.** *In PT-RHM(ρ), give a learner P_{lab} labelled examples, unlimited unlabelled data, and* all *rules except the parent of each fresh root tuple, including which root slots are fresh. Then its expected root error is at least*

$$\frac{\mathbb{E}\big[K_L - \max_a k_{L,a}\big]}{vm} - \frac{P_{\text{lab}}}{vm} \quad \ge \quad (1-\rho) - \frac{1}{v} - \frac{P_{\text{lab}}}{vm}.$$

Proof. A test root uses a fresh slot with probability $K_L/(vm)$. The fresh tuples are assigned to the fresh slots uniformly at random. Unlabelled data carry no information about that assignment: every slot is used with probability $1/(vm)$, and the subtree below a root tuple depends only on the tuple. Given $U \le P_{\text{lab}}$ labelled fresh tuples, an unlabelled fresh tuple has parent a with probability $(k_{L,a} - \ell_a)/(K_L - U)$, where ℓ_a counts the labelled ones in class a. The success probability on fresh roots is therefore at most $U/K_L + \max_a k_{L,a}/K_L$. Multiply by $K_L/(vm)$ and take expectations. The last inequality uses $\max_a k_{L,a} \le m$ and $\mathbb{E}K_L = (1-\rho)vm$. □

At $\rho = 0$ this is Theorem 2. The floor is linear in the tying deficit $1 - \rho$ and does not depend on L.

Lemma 12 (depth–frequency cancellation)**.** *Let $x_h = s^{L-h}$ be the number of height-h nodes, and suppose a slot at height h is used by each node with probability $q_h \in [\gamma, \Gamma]/(vm)$. Call a fresh slot* unlearned *if it occurred fewer than N_0 times in P training examples. Then, for every slot index (a, j) and every L,*

$$\sum_{h=2}^{L-1} x_h q_h \, P\big(\textit{slot } (a,j) \textit{ unlearned at height } h\big) \;\le\; C\,\frac{\Gamma}{\gamma}\,\frac{N_0+1}{P}.$$

Proof. Conditioned on the latents at height h, slot choices are independent. The count is then a sum of Px_h independent indicators with mean Px_hq_h, so $P(\text{unlearned}) \le 1$ if $Px_hq_h < 2N_0$, and

$P(\text{unlearned}) \le e^{-Px_hq_h/8}$ otherwise (Chernoff). The first set of heights forms a geometric series in x_h (ratio s) capped at $2N_0/(\gamma P)\cdot vm$, contributing at most $\frac{s}{s-1}\frac{2N_0\Gamma}{\gamma P}$. The second set contributes $\sum x_hq_he^{-Px_hq_h/8} \le \frac{C\Gamma}{\gamma P}$, again geometrically. □

This is the heart of the matter. *The levels where a deviation is rarely seen are exactly the levels where it rarely matters.* The number of nodes x_h controls both how much data there is for learning a height-h deviation and how often a test tree hits it, so depth drops out.

Theorem 13 (the learner reaches the floor; no depth dependence)**.** *Suppose the conditions of Theorem 4 hold at the bottom, structural naming is correct (Prop. 14), and at every height $h < L$ the sibling-context profiles of distinct classes are Δ_{ctx}-separated, with a constant fraction of usable contexts. Suppose also that every class keeps at least one base rule at every height, which holds with probability at least $1-vL(1-\rho)^m$; capacity-constrained EM removes this condition in practice. Take $N_0 = C\log(vmL/\delta)/\Delta_{\text{ctx}}^2$ and $P \ge Cvm\log(vm/\delta)$. Then the expected root error of the learner, with zero labels, is at most*

$$\underbrace{\frac{\mathbb{E}\big[K_L - \max_a k_{L,a}\big]}{vm}}_{\text{floor (Thm. 11)}} + (1-\rho)\,C\,\frac{\Gamma}{\gamma}\,\frac{vm\,(N_0+1)}{P} \;+\; \delta.$$

Proof. Take a union bound over the nodes of a test tree. A node at height $2 \le h < L$ errs only if it uses an unlearned fresh slot, or a learned one that was misassigned (total probability $\le \delta$ by the choice of N_0 and a union bound over slots and heights). Each slot index is fresh at a given height with probability $1-\rho$. Summing Lemma 12 over the vm slot indices gives the middle term. Tied slots are parsed by the base grammar. At the root, once $P \ge Cvm\log(vm/\delta)$ every kept root slot has been observed, so elimination knows $k_{L,a}$ and attains the floor. □

Theorems 11 and 13 match: the zero-label learner is Bayes-optimal up to an $O\big((1-\rho)vm\log(vmL)/P\big)$ term, and *nothing depends on depth.* At $\rho = 1$ both terms vanish and the result of §3–4 is recovered. Tying buys a reduction of the unavoidable error from ≈ 1 to $\approx (1-\rho)$, and the cost of learning the deviations is spread over the levels exactly as their usage is.

Proposition 14 (structural naming)**.** *Cluster the height-2 id-tuples by their own sibling context to obtain the level-2 synonym partition B. For a candidate naming π, let A_π group two id-tuples iff their images under π are base synonyms. Let $S(\pi)$ be the number of pairs grouped together by both. For the true naming π_0, $S(\pi_0) = \sum_a \kappa_a(\kappa_a - 1)$, where κ_a counts the kept slots of class a at height 2. For $\tau = \pi\pi_0^{-1} \ne \mathrm{id}$ moving the symbol set M: pairs of tuples that avoid M contribute identically to $S(\pi)$ and $S(\pi_0)$. Among pairs touching M, π_0 keeps every pair of kept synonyms (a fraction $\approx \rho^2$ of them), whereas π keeps only pairs whose τ-images happen to be base synonyms. For a random grammar that is a fraction $O(m/(vm)) = O(1/v)$ of them. Hence $S(\pi_0) > S(\pi)$ for all $\tau \ne \mathrm{id}$ whenever ρ^2 exceeds $O(1/v)$ plus fluctuations.*

Structural naming compares *partitions* across scales, not merely valid sets, so it tolerates fresh rules. It also works if fresh rules reuse valid tuples. By contrast, the validity naming of §4 needs $\rho > f$ on average. With $v = 8$ and $\rho = 0.8$ it named 2 of 6 grammars correctly and structural naming 6 of 6; at $\rho = 0.6$ the counts were 0/6 and 6/6 (`verify/test_pt.py`).

Reading Table 4. Each entry is (learner's zero-label error)/(exact floor for the same four grammars). (i) For $0.6 \le \rho \le 0.9$, the learner reaches the floor at $P = 1024$ at *every* depth, e.g.

Table 4: PT-RHM(ρ), v=8, m=4, s=2: zero-label root error of the learner / exact floor of Theorem 11 for the same grammars (means over 4 grammars), by depth L and number of unlabelled examples P. Code: `verify/ptrhm.py`, `verify/exp_pt.py`.

	$P = 256$			$P = 1024$		
ρ	$L = 6$	$L = 8$	$L = 10$	$L = 6$	$L = 8$	$L = 10$
0.9	0.024/0.023	0.081/0.062	0.072/0.047	0.024/0.023	0.067/0.062	0.047/0.047
0.8	0.167/0.117	0.225/0.172	0.169/0.125	0.115/0.117	0.168/0.172	0.127/0.125
0.7	0.395/0.258	0.323/0.188	0.386/0.227	0.270/0.258	0.192/0.188	0.300/0.227
0.6	0.561/0.320	0.608/0.289	0.513/0.328	0.315/0.320	0.290/0.289	0.330/0.328
0.5	0.857/0.438	0.911/0.453	0.964/0.391	0.680/0.438	0.706/0.453	0.655/0.391

0.290/0.289 at $\rho = 0.6$, $L = 8$. The error is linear in the tying deficit and flat in L, as Theorems 11 and 13 predict. The single exception ($\rho = 0.7$, $L = 10$: 0.300/0.227) is one grammar at the largest depth we could afford. (ii) At $P = 256$ the excess over the floor is the $O((1-\rho)vm/P)$ learning term, and it grows as ρ falls, as predicted. (iii) *Breakdown near* $\rho = 0.5$. The bottom partition was still exact in all grammars; the failures have two identified causes. First, structural naming failed in 1 of 4 grammars, because its margin shrinks like ρ^2. Second, when *two* classes keep none of their base rules at the same height, neither has an anchor there, and capacity-constrained assignment cannot tell their deviations apart. For $v = 8$, $m = 4$ and 6 heights, this happens with probability 0.41 at $\rho = 0.5$, 0.10 at 0.6 and 0.01 at 0.7 (Monte Carlo), matching where the learner breaks. This is a limit of the learner, not of the information: exchanging two anchorless classes breaks validity one level up, so cross-level consistency should restore identifiability. We leave that open. (iv) Development note: a first version assigned deviations without capacities. It stalled far above the floor at $\rho = 0.6$, because a class with zero kept rules at some height had its deviations misassigned and the errors cascaded upward. The results with that learner are kept in `verify/exp_pt_results_v1_nocapacity.jsonl`.

7 Continuous check: the linear-Gaussian self-similar tree

Are the gains above an artifact of finite alphabets, exact parsing and combinatorics? We repeat the program for a continuous process.

Definition 15 (LG tree)**.** $z_\epsilon \sim N(0, \Sigma_0)$, and every node's children are $z_{wi} = A_i^{(h)} z_w + \varepsilon_{wi}$ with $\varepsilon \sim N(0, \Psi)$, $i = 1, \ldots, n$. The leaves $x \in \mathbb{R}^d$ are observed. The task is to predict the root vector (score: R^2). *Tied*: $A^{(h)} = A$ at every height. *Untied*: independent maps per height.

Dictionary. Belief propagation becomes the upward Kalman pass, which is a linear recursive grid; its height-dependent gains converge, so it is asymptotically tied. Context clustering becomes the method of moments on sibling cross-covariances. Naming becomes fixing the basis (the gauge) of the latent space. Aut(D) becomes the residual symmetry of the Gaussian law.

Theorem 16 (bottom moments are correlation-free)**.** *Let $n \geq 3$ and let $\hat{C}_{jk}$ be the empirical cross-covariance of children j, k over the N_1 bottom grids. Conditioned on the height-1 latents, $\hat{C}_{jk} = A_j \hat{S} A_k^\top + E_{jk}$, where $\hat{S}$ is the* realized *second moment of the height-1 latents and $\mathbb{E}[\|E_{jk}\|^2 \mid \cdot] = O(1/N_1)$. Hence*

$$\hat{C}_{jk} \hat{C}_{1k}^{-1} = A_j A_1^{-1} + O_P\big(N_1^{-1/2}\big) \qquad (k \notin \{1, j\}),$$

with no dependence on the correlations inside a tree, which all enter through $\hat{S}$ and cancel.

Proof. $x_{wj} = A_j z_w + \varepsilon_{wj}$. The cross terms $A_j \frac{1}{N} \sum z \varepsilon_k^\top$ etc. are conditionally centred with variance $O(1/N_1)$. The ratio cancels $\hat{S}$ exactly, and the error is controlled by $\|(A_1 \hat{S} A_k^\top)^{-1}\|$. □

This is the exact analogue of Theorem 4 and srg.pdf Thm. 5.6: the realized latent covariance plays the role of the realized mixture weights.

Theorem 17 (fixing the basis by self-similarity)**.** *Let $B_j = A_j A_1^{-1}$ (bottom) and $D_j = A_1 B_j A_1^{-1}$ (the same ratios at height 2, estimable from posterior means of the height-1 latents after correcting for their known shrinkage). The solutions X of $XB_j = D_j X$ for all j are exactly $X = A_1 Y$ with Y in the commutant of $\{B_j\}$. If the B_j generate the full matrix algebra (generic; Burnside), $X = cA_1$. Noise consistency ($A_1 \Psi A_1^\top$ must equal the height-2 noise in the learned coordinates) forces $c^2 = 1$. The residual ± 1 is a true symmetry of the model when Σ_0 is symmetric: flipping every latent leaves the leaves' law unchanged. One label, or any asymmetric noise, resolves it.*

Proof. Substitute $X = A_1 Y$: $A_1 Y B_j = A_1 B_j Y$, i.e. $Y B_j = B_j Y$. □

Theorem 18 (untied: labels are unavoidable)**.** *In the untied model with isotropic Σ_0, let the top maps be $\tilde{A}_i U$ with U a Haar rotation independent of everything else. The law of the leaves does not depend on U. A learner that knows everything except U and has P labelled roots satisfies*

$$R^2 \;\le\; \frac{1}{\operatorname{tr} \Sigma_0} \sum_{k \le P} \lambda_k\big(\operatorname{Cov} \mathbb{E}[z_\epsilon \mid x]\big),$$

which is at most $\min(1, P/d)$ times the oracle's R^2 in the isotropic case, and ≤ 0 for $P = 0$.

Proof. $U z_\epsilon \sim N(0, \Sigma_0)$, so the leaves are independent of U. The labels reveal U at most on the span of the P labelled posterior means. On the orthogonal complement, the conditional law of U is invariant under rotations, so the best prediction of that component is 0. □

Learner and a second use of self-similarity. Moments at the bottom, then the Sylvester system at height 2 for the basis, then the Kalman pass with $\hat{A}$ at every height. The root prior Σ_0 is a top-level quantity: estimated from the root posteriors, it has one sample per tree and was badly wrong for few trees. Under *stationarity*, i.e. self-similarity of the marginals, where the root is just another node, Σ_0 equals the height-1 latent covariance, which the bottom estimates from thousands of grids. This fixed it.

Table 5: LG tree, $d = 3$, $n = 3$, $A_i = 0.75\, Q_i$ (Q_i random orthogonal), $\Psi = (1 - 0.75^2) I$, $\Sigma_0 = I$ (oracle $R^2 \approx 0.60$). Zero-label learner (sign-invariant R^2): fraction of 8 random models within 0.05 of the oracle, by depth L and number of unlabelled trees U. Code: `verify/lgtree.py`, `verify/exp_lg.py`.

L	U=1	2	4	8	16	32	64	U^*
4	0	0	0	.13	0	.50	.63	> 64
5	.13	0	0	.13	.50	.75	.88	32
6	0	.13	.25	.75	.75	1	1	8
7	.25	.63	.63	.88	.88	1	1	8
8	.38	.75	1	1	1	1	1	2
9	.75	1	1	1	1	–	–	**1**

Table 6: Untied LG tree ($L = 6$ / $L = 8$; oracle $R^2 = 0.61$). The zero-label learner fails (median $R^2 = -0.22$ / -0.12). A learner told everything except the top rotation fits it by orthogonal Procrustes from P labels; mean R^2 over 8 models.

P	1	2	3	4	8	16
$L = 6$	-0.06	0.05	0.34	0.30	0.35	0.59
$L = 8$	-0.42	0.06	0.26	0.42	0.56	0.57

Table 7: Backprop with root labels on the tied LG tree: median gap to the oracle R^2 (2 models; Adam, 3000 full-batch steps), for a tied linear recursive grid and an untied linear network. The run was stopped before tied $L = 8$, $P = 4096$ and untied $L = 8$ (compute cost).

P	8	16	32	64	128	256	512	1024	4096
tied, $L = 4$	14.1	4.35	0.81	0.12	0.19	0.11	0.17	0.06	0.04
untied, $L = 4$	0.96	1.38	5.76	0.94	0.15	0.07	**0.03**	0.01	0.00
tied, $L = 6$	6.57	3.73	1.18	0.18	0.13	**0.02**	0.01	0.00	0.07
untied, $L = 6$	0.95	1.30	2.09	3.61	0.55	0.15	0.05	**0.02**	0.00
tied, $L = 8$	5.85	7.93	1.72	0.76	0.19	0.09	0.11	0.18	–

What transfers and what does not. *Transfers:* zero-label learning of the shared map from the bottom level; cancellation of within-tree correlations (Thm. 16); unlabelled examples needed falling with depth, down to *one tree* at $L = 9$ (Table 5; the relevant count is again the total number of bottom grids, a few thousand); naming by self-similarity (Thm. 17); and the necessity of labels without tying (Thm. 18, Table 6). *Differs:* (i) Backprop's label cost no longer grows exponentially with depth. In the linear model a few hundred to a thousand labels suffice at $L = 4$–6, and the tied network's difficulties at $L = 8$ look like optimization, not statistics (Table 7). The exponential label cost in the discrete RHM is therefore a property of *combinatorial* structure (discovering synonyms from label correlations), not of self-similarity. What remains is zero labels against hundreds, with data per example that grows with depth. (ii) The residual symmetry is always ± 1 for Gaussian models, where for discrete grammars it was generically trivial. (iii) Continuous inference is sensitive to parameter errors accumulated over many levels, and the root prior needs the stationarity form of self-similarity. *Not yet tested:* nonlinear continuous models, where identification needs more than second moments (the theory is developed in §8), and continuous partial tying. There every child carries information about its parent, so we conjecture a floor $\approx (1-\rho)^n$ for per-position tying instead of $(1-\rho)$.

8 Beyond symbols: noisy and continuous tied processes

The learner of §3–4 has three steps: *context* (Lemma 3: a bottom tuple's sibling context depends only on its parent), *concentration* (Theorem 4: given the latents above, bottom grids are independent, so only their total number matters), and *naming* (Theorem 7, Prop. 14: consistency across scales fixes the labels). None of the three uses discreteness or the absence of noise in an essential way. This section proves each step for an arbitrary tied tree process. The outcome: (a) what context identifies is the grid's *belief* about its parent, i.e. the upward belief-propagation message, so the state space and the cell are dictated by the theory (Theorem 21); (b) depth adds data exactly as before: only the total number of bottom grids enters, now at a nonparametric rate (Theorem 26); (c) naming is identified up to the automorphisms of the kernel (Theorem 28), and additive Gaussian

noise cuts these down to a finite-dimensional group of isometries that commute with the "sibling maps" (Proposition 29), which recovers the ± 1 of Theorem 17.

8.1 Tied tree processes

Definition 19 (tied tree process)**.** Let $\mathcal{Z}$ be a standard Borel space and K a Markov kernel from $\mathcal{Z}$ to $\mathcal{Z}^s$. If $K(\cdot \mid z)$ has a density $k(t \mid z)$ with respect to $\lambda^{\otimes s}$ for a σ-finite λ on $\mathcal{Z}$, we call K *dominated*. Draw the root $z_\epsilon \sim \pi$; every internal node w draws its children's tuple $C_w = (z_{w1}, \dots, z_{ws}) \sim K(\cdot \mid z_w)$, independently given z_w, with the *same* K at every height. The leaves are observed. Latents and leaves take values in the same space $\mathcal{Z}$.

Examples: the tied RHM ($\mathcal{Z} = V$, $k(t \mid a) = \mathbf{1}[t \in R_a]/m$); a *soft grammar* (any k on V^s); the LG tree of §7 ($k(t \mid z) = \prod_i N(t_i; A_i z, \Psi)$); and the *nonlinear Gaussian tree* $k(t \mid z) = \prod_i N(t_i; \varphi_i(z), \Psi)$ with arbitrary measurable φ_i.

As in §3, condition on $\mathcal{H}_2$ (all latents at heights ≥ 2), fix ordered positions $i \neq j$, and let u range over the n_2 height-2 nodes; write $t_{ui} = C_{ui}$ for the tuple of the height-1 grid ui. Let $K_{ij}(\cdot \mid z)$ be the joint law of the i-th and j-th coordinates of a draw from $K(\cdot \mid z)$. Define the realized objects

$$J^n(da, db) = \frac{1}{n_2} \sum_u K_{ij}(da, db \mid z_u), \qquad J^n(da, db) = J_1^n(da)\, \nu^n(db \mid a),$$

(the realized law of a sibling pair of height-1 latents, its first marginal, and its disintegration). Let $m^n(dt) = \int J_1^n(da) K(dt \mid a)$ be the realized tuple law, and let the *belief* β_t^n be the regular conditional law of the parent a given the tuple t under the joint law $J_1^n(da)K(dt \mid a)$; it exists because $\mathcal{Z}$ is standard Borel. If K is dominated, then $m^n(dt) = Z_t^n\, \lambda^{\otimes s}(dt)$ with $Z_t^n = \int k(t \mid a) J_1^n(da)$, and

$$\beta_t^n(da) = \frac{k(t \mid a)\, J_1^n(da)}{Z_t^n}.$$

Finally define the linear *context operator* Λ_n, which maps a finite signed measure β on $\mathcal{Z}$ to $\Lambda_n \beta = \int \beta(da) \int \nu^n(db \mid a) K(\cdot \mid b)$, a signed measure on $\mathcal{Z}^s$, and put $\mu_t^n = \Lambda_n \beta_t^n$.

Lemma 20 (pooled context law)**.** *For every measurable $A \subseteq \mathcal{Z}^s$ and $\psi : \mathcal{Z}^s \to [0, 1]$,*

$$\mathbb{E}\Big[\sum_u \mathbf{1}[t_{ui} \in A]\, \psi(t_{uj}) \;\Big|\; \mathcal{H}_2 \Big] = n_2 \int_A \Big(\int \psi \, d\mu_t^n \Big)\, m^n(dt).$$

So μ_t^n is the pooled conditional law of the sibling tuple given the tuple t: the population object that the empirical context $\hat{\mu}_t$ of §3 estimates. For the RHM it is the μ_t^n of Lemma 3.

Proof. Given $\mathcal{H}_2$, grid u draws $(z_{ui}, z_{uj}) \sim K_{ij}(\cdot \mid z_u)$ and then $t_{ui} \sim K(\cdot \mid z_{ui})$, $t_{uj} \sim K(\cdot \mid z_{uj})$ independently. Hence the left side equals $\sum_u \iint K_{ij}(da, db \mid z_u) K(A \mid a) \int \psi \, dK(\cdot \mid b) = n_2 \int J_1^n(da)\, K(A \mid a)\, F(a)$ with $F(a) = \int \nu^n(db \mid a) \int \psi \, dK(\cdot \mid b)$. By the definition of β_t^n, $\int J_1^n(da) K(A \mid a) F(a) = \int_A m^n(dt) \int \beta_t^n(da) F(a) = \int_A \mu_t^n(\psi)\, m^n(dt)$. For the RHM, $\beta_t^n = \delta_{g(t)}$ and $K(\cdot \mid b) = \mathrm{Unif}(R_b)$. □

8.2 Context identifies beliefs: a general statistical Myhill–Nerode theorem

For a dominated kernel, call two tuples *likelihood-equivalent*, $t \equiv t'$, if $k(t \mid \cdot) = c\, k(t' \mid \cdot)$ J_1^n-almost everywhere for a constant $c > 0$. This is the classical minimal-sufficiency relation of a tuple for its parent.

Theorem 21 (context identifies beliefs)**.** *1. $\mu_t^n = \Lambda_n \beta_t^n$: the context law depends on the tuple only through its belief about the parent.*

2. (K dominated.) If $t \equiv t'$ then $\beta_t^n = \beta_{t'}^n$ and $\mu_t^n = \mu_{t'}^n$, for every *realized histogram.*

3. (K dominated.) If Λ_n is injective on finite signed measures absolutely continuous with respect to J_1^n, then

$$\mu_t^n = \mu_{t'}^n \iff \beta_t^n = \beta_{t'}^n \iff t \equiv t'.$$

The context classes are then exactly the classes of the minimal sufficient statistic of a tuple for its parent. Equivalently, they are the distinct normalized height-1 upward messages of belief propagation (srg.pdf, Thm. 5.2), and the context map $t \mapsto \mu_t^n$ is that message seen through the injective linear map Λ_n. The classes depend on the realized histogram only through the support of J_1^n.

4. If the kernel is parent-determined *(there is a measurable $g : \mathcal{Z}^s \to \mathcal{Z}$ with $K(g^{-1}(a) \mid a) = 1$ for every a; dominated or not), every belief is a point mass $\delta_{g(t)}$, and $\mu_t^n = \mu_{t'}^n \iff g(t) = g(t')$ holds as soon as $a \mapsto \Lambda_n \delta_a$ is injective.*

Proof. (1) is the definition, justified by Lemma 20. (2) The constant c cancels in the normalization of β_t^n. (3) If $\mu_t^n = \mu_{t'}^n$ then $\Lambda_n(\beta_t^n - \beta_{t'}^n) = 0$, so $\beta_t^n = \beta_{t'}^n$, i.e. $k(t \mid a)/Z_t^n = k(t' \mid a)/Z_{t'}^n$ for J_1^n-a.e. a, which is $t \equiv t'$ with $c = Z_t^n / Z_{t'}^n$. The converse is (2). (4) The joint law $J_1^n(da)K(dt \mid a)$ is concentrated on $\{a = g(t)\}$, so $\beta_t^n = \delta_{g(t)}$ for m^n-a.e. t, and $\mu_t^n = \Lambda_n \delta_{g(t)}$. □

Remark 22 (the injectivity condition)**.** Λ_n is the composition of $\beta \mapsto \int \beta(da)\nu^n(\cdot \mid a)$ (the realized sibling channel between latents) and $\gamma \mapsto \int \gamma(db)K(\cdot \mid b)$ (the kernel as a map from mixing laws to tuple laws). It is injective if both are: the realized sibling channel is invertible (for finite $\mathcal{Z}$: the matrix ν^n is invertible), and K is *mixture-identifiable* (a tuple law determines its mixing law over parents; automatic for disjoint supports). In the parent-determined case (4) asks only that distinct parents have distinct context laws. For the RHM, where the $K(\cdot \mid b)$ have disjoint supports, $\|\Lambda_n \delta_{a_1} - \Lambda_n \delta_{a_2}\|_1 = \|\nu^n(\cdot|a_1) - \nu^n(\cdot|a_2)\|_1$, so this is exactly the condition $\Delta(n) > 0$ of Theorem 4.

Corollary 23 (three regimes)**.** *1.* Parent-determined kernels *(the RHM; also its continuous form $K(\cdot \mid z) = \frac{1}{m}\sum_{r=1}^m \delta_{\Phi_r(z)}$ with injective Φ_r and pairwise disjoint images). The classes are the fibres of the parent map g: the synonyms. If $a \mapsto \Lambda_n \delta_a$ is injective, the context map $t \mapsto \Lambda_n \delta_{g(t)}$ is an injective function of the parent, so it recovers the parent up to the bijection $z \mapsto \Lambda_n \delta_z$. This bijection is the* gauge *that naming must remove.*

2. Exponential-family kernels $k(t \mid a) = \eta_0(t) \exp\big(\eta(a)^\top S(t) - A(a)\big)$. *Then $t \equiv t'$ iff $S(t) - S(t') \perp \operatorname{span}\{\eta(a) - \eta(a') : a, a' \in \operatorname{supp} J_1^n\}$. For the LG tree, $\eta(a) = (A_1 a, \dots, A_s a)$ spans a d-dimensional space, and the classes are the level sets of the d-dimensional statistic $\sum_i A_i^\top \Psi^{-1} t_i$ (the Kalman update): a linear tree compresses a tuple to d numbers. For the nonlinear Gaussian tree, $\eta(a) = (\varphi_1(a), \dots, \varphi_s(a))$. If these vectors affinely span $\mathbb{R}^{sd}$ as a ranges over the support, which is typical for nonlinear φ_i, then no two distinct tuples are equivalent.*

3. Soft grammars. *Likewise, for generic k all tuples have distinct likelihood vectors.*

In cases (2)–(3) nothing collapses into finitely many classes. What the bottom level identifies is the whole belief map $t \mapsto \beta_t^n$, up to the linear map Λ_n.

Proof. (1) is Theorem 21(4). (2) $k(t \mid a)/k(t' \mid a) = \frac{\eta_0(t)}{\eta_0(t')} \exp\big(\eta(a)^\top (S(t) - S(t'))\big)$ is constant in a iff $(\eta(a) - \eta(a'))^\top (S(t) - S(t')) = 0$ for all a, a'. For the Gaussian trees, $S(t) = (\Psi^{-1} t_1, \dots, \Psi^{-1} t_s)$, and $\sum_i (A_i a)^\top \Psi^{-1} t_i = a^\top \sum_i A_i^\top \Psi^{-1} t_i$. □

Remark 24 (the state is a belief, and it is the same type as a leaf)**.** Theorem 21 says what the outside state of a grid must be for context to act on it: its normalized likelihood for the parent, i.e. its belief. A leaf's belief is the point mass δ_x. So leaves and grids have states of the *same type*, namely probability measures on $\mathcal{Z}$, in every regime. This is the precise form of "latents are the same kind of symbol as leaves", which §4 needed. The cell that maps the children's beliefs to the parent's is the sum–product map of K (srg.pdf, Thm. 5.2). Once K is known, nothing else remains to be learned: no representation and no auxiliary objective. Only in the parent-determined regime may the state be taken to be a point; there beliefs *are* points. With genuine noise a point-valued state discards the spread of the belief. That spread is exactly what has to be combined with the evidence from the parent's other children.

8.3 Depth adds data, in general

The proof of Theorem 4 used only one fact: conditioned on $\mathcal{H}_2$, the bottom grids of different height-2 nodes are independent. That fact holds for every tied tree process. In the continuous case the context law of a single tuple is not observed; it is estimated by pooling the tuples that fall into a small bin, and the price is a bias that we control through the beliefs.

Lemma 25 (belief diameter)**.** *If* $\sup_a |\log k(t \mid a) - \log k(t' \mid a)| \le \varepsilon$*, then* $\|\beta^n_t - \beta^n_{t'}\|_{\mathrm{TV}} \le 1 - e^{-2\varepsilon} \le 2\varepsilon$ *for every realized histogram.*

Proof. The ratio $d\beta^n_t / d\beta^n_{t'} = \frac{k(t|a)}{k(t'|a)} \cdot \frac{Z^n_{t'}}{Z^n_t}$ lies in $[e^{-2\varepsilon}, e^{2\varepsilon}]$, since both factors lie in $[e^{-\varepsilon}, e^{\varepsilon}]$. Hence $\|\beta^n_t - \beta^n_{t'}\|_{\mathrm{TV}} = \int (1 - \frac{d\beta_t}{d\beta_{t'}})_+ \, d\beta_{t'} \le 1 - e^{-2\varepsilon}$. □

Theorem 26 (context estimation; only the total number of bottom grids enters)**.** *Condition on* $\mathcal{H}_2$*. Let* $\mathfrak{B}$ *be a finite family of disjoint measurable bins in* $\mathcal{Z}^s$ *and* Ψ_0 *a finite family of test functions* $\mathcal{Z}^s \to [0,1]$*. For* $B \in \mathfrak{B}$*, let* $N_B = \#\{u : t_{ui} \in B\}$*, let* $\hat{\mu}_B(\psi) = N_B^{-1} \sum_{u : t_{ui} \in B} \psi(t_{uj})$*, and let* $\omega(B) = \sup_{t,t' \in B,\ \psi \in \Psi_0} |\mu^n_t(\psi) - \mu^n_{t'}(\psi)|$ *be the context diameter of* B*. Since* Λ_n *maps probability measures to probability measures,* $\omega(B) \le \sup_{t,t' \in B} \|\beta^n_t - \beta^n_{t'}\|_{\mathrm{TV}}$*. There are absolute constants* C, C' *such that, with probability* $\ge 1 - \delta$*, simultaneously for all bins with* $n_2 m^n(B) \ge C \log(2|\mathfrak{B}||\Psi_0|/\delta)$*, all* $\psi \in \Psi_0$ *and all* $t \in B$*,*

$$\left|\hat{\mu}_B(\psi) - \mu^n_t(\psi)\right| \;\le\; \omega(B) + C' \sqrt{\frac{\log(2|\mathfrak{B}||\Psi_0|/\delta)}{n_2 \, m^n(B)}}.$$

Proof. Let $X_u = \mathbf{1}[t_{ui} \in B]$, $Y_u = X_u \psi(t_{uj})$, and $\bar{\mu}_B(\psi) = \int_B \mu^n_t(\psi) \, m^n(dt) / m^n(B)$. By Lemma 20, $\mathbb{E} \sum_u (Y_u - \bar{\mu}_B X_u) = 0$. The summands are independent given $\mathcal{H}_2$ and lie in $[-1, 1]$, with total variance at most $\sum_u \mathbb{E} X_u = n_2 m^n(B)$. Bernstein's inequality for the numerator and a Chernoff bound $N_B \ge n_2 m^n(B)/2$ give $|\hat{\mu}_B(\psi) - \bar{\mu}_B(\psi)| = |\sum_u (Y_u - \bar{\mu}_B X_u)| / N_B \le C' \sqrt{\log(\cdot)/(n_2 m^n(B))}$. Next, $\bar{\mu}_B(\psi)$ is an average of $\mu^n_{t'}(\psi)$ over $t' \in B$, so $|\bar{\mu}_B(\psi) - \mu^n_t(\psi)| \le \omega(B)$. Finish with a union bound over bins and test functions. □

Corollary 27 (one example suffices, continuous case)**.** *Let* K *be dominated. Suppose the tuples fall in a region covered by* $N(\eta)$ *bins of diameter* η *(in a metric on* $\mathcal{Z}^s$*). Suppose also that* $\log k(\cdot \mid a)$ *is* G*-Lipschitz there, uniformly in* a*, and that every bin has realized mass* $m^n(B) \ge \gamma / N(\eta)$*. Then every context law is estimated to accuracy* ε *on* Ψ_0*, with probability* $\ge 1 - \delta$*, as soon as*

$$n_2 = P \, s^{L-2} \;\ge\; C \, \frac{N\big(\varepsilon/4G\big) \, \log\big(N(\varepsilon/4G)|\Psi_0|/\delta\big)}{\gamma \, \varepsilon^2}.$$

If the tuples have intrinsic dimension D, then $N(\eta) \asymp \eta^{-D}$ and the requirement is $n_2 \gtrsim \varepsilon^{-(D+2)}$ up to logarithms. It is a requirement on n_2 alone. So the number of examples needed falls like $s^{-(L-2)}$, and one example suffices once s^{L-2} exceeds it.

Proof. Lemma 25 gives $\omega(B) \le 2G\eta = \varepsilon/2$ for $\eta = \varepsilon/4G$. Make the square-root term of Theorem 26 at most $\varepsilon/2$. □

For finite $\mathcal{Z}$ take singleton bins: then $\omega = 0$, $N = |D|$, and this is Theorem 4. For continuous parent-determined kernels the Lipschitz condition is replaced by continuity of $a \mapsto \Lambda_n \delta_a(\psi)$ along the parent map g. Continuity changes the *exponent* (nonparametric, $D + 2$), not the depth law. Under a parametric kernel family (such as the exponential families of Corollary 23), bins are replaced by moments of S and the rate is again $n_2^{-1/2}$, as in Theorem 16. Correlations inside a tree enter only through the realized quantities J^n, m^n and Λ_n, exactly as before.

8.4 Naming as a kernel automorphism

Naming is a statement about the *population* objects that the lower two levels deliver. Let $\mathcal{Z}_1$ and $\mathcal{Z}_2$ be the sets of values taken by the height-1 and height-2 latents, and assume $\mathcal{Z}_2 \subseteq \mathcal{Z}_1$ (every value that occurs as a height-2 parent also occurs at height 1; this is automatic under stationarity, and for the RHM it holds with high probability).

(N) The learner knows, for unknown bijections $h : \mathcal{Y} \to \mathcal{Z}_1$ and $h_2 : \mathcal{X} \to \mathcal{Z}_2$: (a) the bottom kernel in its own coordinates, $P_y = K(\cdot \mid h(y))$ for $y \in \mathcal{Y}$; and (b) the level-2 kernel in those coordinates, $Q_x = (h^{-1})_{\#}^{\otimes s} K(\cdot \mid h_2(x))$ for $x \in \mathcal{X}$, i.e. the law of the coordinates y of a height-2 grid's children given its (arbitrarily labelled) class x.

A *naming* is a candidate $h' : \mathcal{Y} \to \mathcal{Z}$ (injective, measurable). It asserts the kernel $K'(\cdot \mid z) = P_{h'^{-1}(z)}$ and the leaf-type values $h'(y)$ of the height-1 latents. It is *consistent with self-similarity* if every level-2 law, mapped into leaf coordinates, is a member of the asserted kernel: for every x there is $z' \in h'(\mathcal{Y})$ with $h'^{\otimes s}_{\#} Q_x = K'(\cdot \mid z')$.

Theorem 28 (naming by self-similarity, general)**.** *Define the automorphism set of the kernel*

$$\mathrm{Aut}(K) = \Big\{ \tau : \mathcal{Z}_1 \to \mathcal{Z} \text{ injective measurable } : \ \forall z \in \mathcal{Z}_2 \ \exists z'' \in \mathcal{Z}_1, \quad \tau_{\#}^{\otimes s} K(\cdot \mid z) = K(\cdot \mid z'') \Big\}.$$

Under (N), a naming h' is consistent iff $h' = \tau \circ h$ with $\tau \in \mathrm{Aut}(K)$. In particular, the true naming h is identified with zero labels iff $\mathrm{Aut}(K) = \{\mathrm{id}\}$. Otherwise it is identified up to $\mathrm{Aut}(K)$, and labels must decide among the remaining candidates.

Proof. Write $\tau = h' \circ h^{-1}$, so $h' = \tau \circ h$, and $z = h_2(x) \in \mathcal{Z}_2$. Then $h'^{\otimes s}_{\#} Q_x = \tau_{\#}^{\otimes s} K(\cdot \mid z)$, and $K'(\cdot \mid z') = P_{h'^{-1} z'} = K(\cdot \mid \tau^{-1} z')$ with $z'' = \tau^{-1} z'$ ranging over $h(\mathcal{Y}) = \mathcal{Z}_1$ as z' ranges over $h'(\mathcal{Y})$. So consistency says: for every $z \in \mathcal{Z}_2$ there is $z'' \in \mathcal{Z}_1$ with $\tau_{\#}^{\otimes s} K(\cdot \mid z) = K(\cdot \mid z'')$, i.e. $\tau \in \mathrm{Aut}(K)$. Every step is reversible. □

Special cases. For the RHM, $\mathrm{Aut}(K)$ is the set of permutations mapping every synonym block R_a onto a synonym block. That is a subgroup of $\mathrm{Aut}(D)$ (Theorem 7 used only validity, a weaker test), and it is the population version of structural naming (Prop. 14). For a conditionally independent discrete kernel $K(\cdot \mid z) = \bigotimes_i M_i[:, z]$, with τ a permutation matrix R, the condition reads $RM_i = M_i S$ for a permutation S; this is the condition of srg.pdf, Thm. 5.3(2), after rearrangement.
How (N) is delivered. For parent-determined kernels, Theorems 21(4) and 26 applied at height 1 give (a), with $y = \Lambda_n \delta_z$ as the coordinate. Applied at height 2 to the y-tuples, which are again

parent-determined, they give (b). With genuine noise, (a) is *mixture identification* of the bottom tuple law (which parent law produced which tuple law). (b) additionally needs the channel $y \mapsto P_y$ to be injective, so that the law of the height-1 latents in y-coordinates can be recovered from the law of whole height-2 blocks. For the LG tree both hold, and §7 carried them out ("correcting for the known shrinkage"). For bounded nonlinear Gaussian kernels, Theorem 36 below proves (a) and the channel part of (b), and shows that the grouping in (b) is not needed.

Proposition 29 (additive Gaussian noise: naming up to commuting isometries)**.** *Let $\mathcal{Z} = \mathbb{R}^d$ and $K(\cdot \mid z) = \bigotimes_{i=1}^{s} N(\varphi_i(z), \Psi)$ with $\Psi \succ 0$ and measurable φ_i, so that $\mathcal{Z}_1 = \mathbb{R}^d$. Suppose that for some i_0 the set $\varphi_{i_0}(\mathcal{Z}_2)$ affinely spans $\mathbb{R}^d$.*

1. *Every $\tau \in \mathrm{Aut}(K)$ agrees almost everywhere with a Ψ-isometry $T(x) = Ux + b$, $U\Psi U^\top = \Psi$. Moreover, there is a map S with $T\varphi_i(z) = \varphi_i(S(z))$ for all i and all $z \in \mathcal{Z}_2$.*
2. *Conversely, every Ψ-isometry T with such an S lies in $\mathrm{Aut}(K)$.*
3. *If φ_1 is injective, then $\mathrm{Aut}(K)$ is (up to null sets) the group of Ψ-isometries commuting with every* sibling map *$B_i = \varphi_i \circ \varphi_1^{-1}$ on $\varphi_1(\mathcal{Z}_2)$, provided T maps $\varphi_1(\mathcal{Z}_2)$ into $\varphi_1(\mathbb{R}^d)$. This is a closed subgroup of a Lie group of dimension $d(d+1)/2$.*
4. *For the LG tree ($\varphi_i = A_i$ linear) with $\{B_i\}$ generating the full matrix algebra and $\bigcap_i \ker(B_i - I) = 0$: $\mathrm{Aut}(K) = \{\pm\mathrm{id}\}$, which is Theorem 17.*

Proof. (1) Taking the i_0-th marginal, $\tau_\# N(m, \Psi)$ is Gaussian with covariance Ψ for every $m \in M = \varphi_{i_0}(\mathcal{Z}_2)$. Pick affinely independent $m_0, \dots, m_d \in M$, and let m'_k be the corresponding means of the images. Radon–Nikodym derivatives are transported by bijections: $\frac{d\tau_\#\mu}{d\tau_\#\nu}(\tau x) = \frac{d\mu}{d\nu}(x)$. Applied to $\mu = N(m_k, \Psi)$ and $\nu = N(m_0, \Psi)$, this gives, for a.e. x,

$$(m_k - m_0)^\top \Psi^{-1} x = (m'_k - m'_0)^\top \Psi^{-1} \tau(x) + c_k, \qquad k = 1, \dots, d.$$

Stacked, the left sides are $A^\top \Psi^{-1} x$ with $A = [m_k - m_0]_k$ invertible. If $A' = [m'_k - m'_0]_k$ were singular, the right sides would lie in a proper affine subspace for a.e. x, which the left sides do not. So A' is invertible, and $\tau(x) = Ux + b$ a.e., with $U = \Psi A'^{-\top} A^\top \Psi^{-1}$. Then $\tau_\# N(m, \Psi) = N(Um + b, U\Psi U^\top)$, so $U\Psi U^\top = \Psi$. Comparing the means of every marginal gives $T\varphi_i(z) = \varphi_i(S(z))$ with $S(z) = z''$. (2) $T^{\otimes s}$ maps $\bigotimes_i N(\varphi_i(z), \Psi)$ to $\bigotimes_i N(U\varphi_i(z) + b, \Psi) = \bigotimes_i N(\varphi_i(Sz), \Psi) = K(\cdot \mid Sz)$. (3) The first coordinate gives $S = \varphi_1^{-1} \circ T \circ \varphi_1$. Substituting, $T\varphi_i(z) = \varphi_i \varphi_1^{-1} T \varphi_1(z)$, i.e. $TB_i = B_i T$ at every point $\varphi_1(z)$. Commuting isometries form a closed subgroup of the Ψ-isometry group, whose dimension is $d(d-1)/2 + d$. (4) $TB_i = B_i T$ with $T = U \cdot + b$ and linear B_i gives $UB_i = B_i U$ and $(B_i - I)b = 0$. By Burnside, $U = cI$, and $U\Psi U^\top = \Psi$ forces $c = \pm 1$; the kernel condition forces $b = 0$. □

Remark 30 (noise is what makes continuous naming finite)**.** Without noise, $K(\cdot \mid z) = \delta_{(\varphi_i(z))_i}$, the same computation leaves $\mathrm{Aut}(K)$ equal to *all* bijections commuting with the sibling maps. This can be infinite-dimensional. In the noise-free version of the LG tree of Table 5, where the $B_i = Q_i Q_1^{-1}$ are orthogonal, every radial reparametrization $x \mapsto \rho(\|x\|)\, x/\|x\|$ with ρ an increasing bijection of $[0, \infty)$ commutes with every B_i. The Gaussian noise law is a ruler in leaf coordinates: it forces τ to be affine, and reduces the ambiguity to a finite-dimensional group, which is $\{\pm\mathrm{id}\}$ in the linear case and should be trivial for generic nonlinear φ_i. Like $|\mathrm{Aut}(D)|$ in Table 1, $\mathrm{Aut}(K)$ is computable for any given kernel. Noise therefore costs something in Theorem 21 (beliefs instead of points) and in (N) (mixture identification), and pays it back here.

8.5 Hypothesis (N) for the nonlinear Gaussian tree

We now prove (N) for nonlinear Gaussian trees with bounded maps. We also show that the *component* form of (N)(b), with height-2 grids grouped by their parent, is not needed: the height-2 *mixture* already determines the naming up to $\mathrm{Aut}(K)$. Throughout, $\mathbb{B}$ is the closed unit ball of $\mathbb{R}^d$, $\mathbb{B}^\circ$ its interior, and $\sigma(z) = z/(1+\|z\|)$ is a homeomorphism $\mathbb{R}^d \to \mathbb{B}^\circ$.

Definition 31 (bounded nonlinear Gaussian tree)**.** A tied tree process on $\mathcal{Z} = \mathbb{R}^d$ with $s \geq 3$ and $K(\cdot \mid z) = \bigotimes_{i=1}^s N(\varphi_i(z), \Psi)$, where $\Psi \succ 0$ is unknown and $\varphi_i = g_i \circ \sigma$ for continuous maps $g_i : \mathbb{B} \to \mathbb{R}^d$ that are injective on the *closed* ball.

So each φ_i is bounded and injective, and it extends continuously and injectively to the "points at infinity" $\partial\mathbb{B}$. Write $\Phi = (\varphi_1, \ldots, \varphi_s)$, $G = (g_1, \ldots, g_s)$, $\Sigma_s = I_s \otimes \Psi$, and $B_i = g_i \circ g_1^{-1}$ on $g_1(\mathbb{B})$ for the (extended) sibling maps. For $C = \{3, \ldots, s\}$, let the *exposed parents* $E \subseteq \mathbb{B}$ be the points that are the unique maximizer over $\mathbb{B}$ of $p \mapsto \sum_{k \in C} \theta_k^\top g_k(p)$ for some $\theta \in \mathbb{R}^{|C|d}$.

(A) $g_1(E) \cup g_2(E)$ affinely spans $\mathbb{R}^d$.

The *data* are the law of a bottom tuple and the law of a height-2 block (its s^2 leaves), as mixtures over height-1 and height-2 latent laws ρ_1, ρ_2. Each of these is itself a mixture of Gaussians and so has a positive Lebesgue density, whether it is the population law or the law realized given the latents above. Theorem 26 supplies n_2 and n_3 conditionally independent copies of the two.

We need four lemmas. Write $\mathcal{N}_\Sigma = N(0, \Sigma)$, and $\hat{Q}$ for a Fourier transform.

Lemma 32 (the Gaussian part of a compactly supported mixture)**.** *If $Q * \mathcal{N}_\Sigma = Q' * \mathcal{N}_{\Sigma'}$ for compactly supported probability measures Q, Q' on $\mathbb{R}^D$ and $\Sigma, \Sigma' \succeq 0$, then $\Sigma = \Sigma'$ and $Q = Q'$.*

Proof. If Q is supported in a ball of radius R, then $\hat{Q}$ extends to an entire function on $\mathbb{C}^D$ with $|\hat{Q}(\omega)| \leq e^{R|\operatorname{Im}\omega|}$, and likewise $\hat{Q}'$. The identity $\hat{Q}(\omega)e^{-\omega^\top\Sigma\omega/2} = \hat{Q}'(\omega)e^{-\omega^\top\Sigma'\omega/2}$ therefore holds on $\mathbb{C}^D$. Fix $v \in \mathbb{R}^D$ and put $f(w) = \hat{Q}(wv)$, $g(w) = \hat{Q}'(wv)$ for $w \in \mathbb{C}$. Both are entire of order ≤ 1, $f(0) = 1$, and $g = f\,e^{aw^2}$ with $a = v^\top(\Sigma' - \Sigma)v/2$. By Hadamard's factorization, f and g have the same zeros, so $g/f = e^P$ with P a polynomial of degree ≤ 1. Then $aw^2 - P(w)$ is constant, so $a = 0$. As v is arbitrary, $\Sigma = \Sigma'$, and then $\hat{Q} = \hat{Q}'$. □

Lemma 33 (partial deconvolution)**.** *Let R, R' be finite measures on $\mathbb{R}^d \times W$ (W standard Borel), and let $*_1$ denote convolution in the first factor. If $R *_1 \mathcal{N}_\Psi = R' *_1 \mathcal{N}_\Psi$, then $R = R'$.*

Proof. For bounded measurable u on W, the finite signed measures $R_u(dm) = \int u(w)R(dm, dw)$ and R'_u satisfy $R_u * \mathcal{N}_\Psi = R'_u * \mathcal{N}_\Psi$. Divide the Fourier transforms by the nonvanishing $e^{-\omega^\top\Psi\omega/2}$. □

Lemma 34 (a smoothed product has a product mixing law)**.** *If Q is a probability measure on $\mathbb{R}^d \times \mathbb{R}^d$ and $Q * (\mathcal{N}_\Psi \otimes \mathcal{N}_\Psi) = F \otimes F'$, then $Q = Q_1 \otimes Q_2$.*

Proof. $\hat{Q}(\omega_1, \omega_2) = a(\omega_1)b(\omega_2)$ with $a = \hat{F}e^{\omega_1^\top\Psi\omega_1/2}$ and $b = \hat{F}'e^{\omega_2^\top\Psi\omega_2/2}$. Since $a(0)b(0) = \hat{Q}(0,0) = 1$, we get $\hat{Q}(\omega_1, \omega_2) = \hat{Q}(\omega_1, 0)\hat{Q}(0, \omega_2)$. □

Lemma 35 (tilting concentrates on the maximizer)**.** *Let ν be a probability measure on $\mathbb{B}^\circ$ with full support, $f : \mathbb{B} \to \mathbb{R}$ continuous with a unique maximizer p^*, and a_r functions with $\sup|a_r| \leq A$. Then the measures $\nu_r(dp) \propto e^{rf(p)+a_r(p)}\nu(dp)$ converge weakly on $\mathbb{B}$ to δ_{p^*} as $r \to \infty$.*

Proof. Let $U \ni p^*$ be open and $\delta = f(p^*) - \max_{\mathbb{B}\setminus U} f > 0$. The open set $V = U \cap \{f > f(p^*) - \delta/2\}$ contains $p^* \in \overline{\mathbb{B}^\circ}$, so $\nu(V) > 0$. Then $\nu_r(\mathbb{B} \setminus U)/\nu_r(V) \leq e^{2A}e^{-r\delta/2}/\nu(V) \to 0$. □

Theorem 36 ((N) and naming for the bounded nonlinear Gaussian tree)**.** *Under Definition 31:*

1. Noise. *The law of one leaf of a bottom tuple determines* Ψ *and the law of* $\varphi_1(z)$ *over the height-1 latents.*

2. (N)(a). *In the gauge coordinate* $m = \varphi_1(z)$*, the bottom tuple law determines the kernel:* $P_m = N(m, \Psi) \otimes \bigotimes_{i\geq 2} N(B_i(m), \Psi)$ *for every* $m \in g_1(\mathbb{B})$*. Here* $h = \varphi_1^{-1}$*.*

3. (N)(b), channel part. *The law of a height-2 block determines the law* G_2 *of* $(m_1, \dots, m_s) = (\varphi_1(z_1), \dots, \varphi_1(z_s))$*, the height-1 latents of a height-2 grid in gauge coordinates.*

4. Naming. *Let a candidate naming be an injective measurable* $h' : g_1(\mathbb{B}^\circ) \to \mathbb{R}^d$*. It asserts the kernel* $K'(\cdot \mid z) = P_{h'^{-1}(z)}$*, and it is* consistent *if* $h'^{\otimes s}_{\#} G_2 = \int \rho'(dz) K'(\cdot \mid z)$ *for some mixing law* ρ'*. Assume (A). Then* h' *is consistent iff* $\tau = h' \circ \varphi_1$ *agrees almost everywhere with some*

$$T \in \mathrm{Aut}(K) = \big\{T(x) = Ux + b : \, U\Psi U^\top = \Psi, \, \exists S \text{ measurable with } T\varphi_i(z) = \varphi_i(S(z)) \, \forall i, \text{ a.e. } z\big\}.$$

In particular, the conclusion of Theorem 28 holds using only the mixture *over height-2 parents. The naming is identified with zero labels iff* $\mathrm{Aut}(K) = \{\mathrm{id}\}$*, and by Proposition 29(3) this is the group of* Ψ*-isometries commuting with the sibling maps.*

Proof. (1) The first leaf of a bottom tuple has law $(\varphi_{1\#}\rho_1) * \mathcal{N}_\Psi$, and $\varphi_{1\#}\rho_1$ is supported in the compact set $g_1(\mathbb{B})$. Any other model of the same class gives the same form. Apply Lemma 32.

(2) The noise ε_1 of the first leaf is independent of $(z, t_2, \dots, t_s)$, so the tuple law is $R *_1 \mathcal{N}_\Psi$ with R the joint law of $(\varphi_1(z), t_2, \dots, t_s)$. By Lemma 33, R is determined. Its conditional law given $m = \varphi_1(z)$ is $\bigotimes_{i\geq 2} N(\varphi_i\varphi_1^{-1}(m), \Psi)$ for $\varphi_{1\#}\rho_1$-a.e. m, which determines B_i on a dense subset of $g_1(\mathbb{B})$ (the density of ρ_1 is positive), hence everywhere by continuity.

(3) In a height-2 block, child i's first leaf is $\varphi_1(z_i) + \varepsilon_i$, with the s noises independent of each other and of everything else. Deconvolve these s coordinates (Lemma 33, applied s times) and marginalize.

(4) Let $A = \int \rho_2(dz) K(\cdot \mid z)$ be the law of $(z_1, \dots, z_s)$ in leaf coordinates, so $G_2 = \varphi_1^{\otimes s}{}_{\#} A$, and $h'^{\otimes s}_{\#} G_2 = \tau^{\otimes s}_{\#} A$. Since $K'(\cdot \mid \tau(w)) = P_{\varphi_1(w)} = K(\cdot \mid w)$, consistency reads

$$\tau^{\otimes s}_{\#} A = \int \rho''(dw) K(\cdot \mid w) \qquad \big(\rho'' = \tau^{-1}_{\#}\rho'\big). \tag{$*$}$$

If. If $\tau = T$ a.e., then $T_{\#} N(m, \Psi) = N(Tm, \Psi)$ gives $\tau^{\otimes s}_{\#} A = \int \rho_2(dz) \bigotimes_i N(\varphi_i(Sz), \Psi) = \int (S_{\#}\rho_2)(dw) K(\cdot \mid w)$.

Only if, step 1: conditionals. Write P for the common law in $(*)$, with coordinates $X_1, \dots, X_s$ and $X_C = (X_k)_{k\in C}$. Two versions of the conditional law of (X_1, X_2) given $X_C = y$ are available. The right side gives $\int \beta''_y(dw) \, N(\varphi_1(w), \Psi) \otimes N(\varphi_2(w), \Psi)$, where β''_y is the posterior of w given $X_C = y$ under ρ''. The left side gives $(\tau \otimes \tau)_{\#} \int \beta_x(dz) \, N(\varphi_1(z), \Psi) \otimes N(\varphi_2(z), \Psi)$ with $x = \tau^{-1}(y)$, where β_x is the posterior of z given $(z_k)_{k\in C} = x$ under ρ_2. The two agree for y in a set $\mathcal{G}$ of full P_{X_C}-measure. Since the law of $(z_k)_{k\in C}$ has a positive density and $(\tau^{-1})^{\otimes |C|}(\mathcal{G})$ has full measure under it, that set has full Lebesgue measure.

Step 2: extreme conditioning. Fix $p^* \in E$ and a θ for which it is the unique maximizer. Choose $x^{(r)} \in (\tau^{-1})^{\otimes |C|}(\mathcal{G})$ with $\|x^{(r)} - r\Psi\theta\| < 1$. In the coordinate $p = \sigma(z)$, the posterior $\beta_{x^{(r)}}$ is $\sigma_{\#}\rho_2$ tilted by $\exp\big(r \sum_k \theta_k^\top g_k(p) + O(1)\big)$, with the $O(1)$ uniform because the g_k are bounded. By Lemma 35 it converges weakly to δ_{p^*}. The map $p \mapsto (\tau \otimes \tau)_{\#}[N(g_1(p), \Psi) \otimes N(g_2(p), \Psi)]$ is continuous in total variation, since bijections preserve total variation. So the left-side conditionals

converge in total variation to $\tau_{\#}N(g_1(p^*),\Psi)\otimes\tau_{\#}N(g_2(p^*),\Psi)$, a *product*. The right-side posteriors, viewed on the compact $\mathbb{B}$, have a weakly convergent subsequence $\beta'' \to \beta''_\infty$, and the right-side conditionals converge in total variation to $Q * (\mathcal{N}_\Psi \otimes \mathcal{N}_\Psi)$ with $Q = (g_1, g_2)_{\#}\beta''_\infty$.

Step 3: the right side collapses. By Lemma 34, $Q = Q_1 \otimes Q_2$. Its support $\operatorname{supp} Q_1 \times \operatorname{supp} Q_2$ lies in $(g_1, g_2)(\mathbb{B}) = \{(m, B_2(m))\}$, the graph of a bijection, because g_1 and g_2 are injective on $\mathbb{B}$. A product set inside such a graph is a single point, so $Q = \delta_{(g_1(q^*),g_2(q^*))}$, and

$$\tau_{\#}N(g_1(p^*),\Psi) = N(g_1(q^*),\Psi), \qquad \tau_{\#}N(g_2(p^*),\Psi) = N(g_2(q^*),\Psi).$$

Step 4: τ is an isometry. By (A) the means $g_1(p^*), g_2(p^*)$, $p^* \in E$, include $d+1$ affinely independent points. The Radon–Nikodym argument of Proposition 29(1) then gives $\tau = T$ a.e. with $T(x) = Ux + b$ and $U\Psi U^\top = \Psi$.

Step 5: T conjugates the kernel. Now $\tau_{\#}^{\otimes s}A = (T^{\otimes s}\Phi)_{\#}\rho_2 * \mathcal{N}_{\Sigma_s}$, while the right side of $(*)$ is $\Phi_{\#}\rho'' * \mathcal{N}_{\Sigma_s}$. Deconvolving gives $(T^{\otimes s}\Phi)_{\#}\rho_2 = \Phi_{\#}\rho''$. So $T^{\otimes s}\Phi(z) \in \Phi(\mathbb{R}^d)$ for ρ_2-a.e., hence Lebesgue-a.e., z, and $S = \Phi^{-1} \circ T^{\otimes s} \circ \Phi$ is measurable (Φ is injective and continuous) with $T\varphi_i = \varphi_i \circ S$ a.e. □

Remark 37 (how restrictive (A) is)**.** E contains G_C^{-1} of every exposed point of $\operatorname{conv} G_C(\mathbb{B})$, where $G_C = (g_k)_{k\in C}$ is injective. The support function of a compact convex set is differentiable almost everywhere, and at such directions the maximizing face is a single exposed point. So E is nonempty, and (A) fails only if g_1 and g_2 send all exposed parents into one hyperplane. For $d = 1$, the injective g_k are strictly monotone, $E = \{-1, +1\}$ (the two points at infinity), and (A) holds automatically. There $\operatorname{Aut}(K) \subseteq \{x \mapsto \pm x + b\}$, and it is trivial unless every g_i satisfies $g_i(Sp) = \pm g_i(p) + b$ for one common S.

Remark 38 (what the theorem does and does not say)**.** It is an *identification* result: the bottom two levels' laws determine the kernel and its naming (up to $\operatorname{Aut}(K)$), and a single tree supplies those laws through n_2 and n_3 conditionally independent copies (Theorem 26). The proof goes through deconvolution, which is statistically ill-posed as a nonparametric procedure. The learner it licenses is not a deconvolution but maximum likelihood (EM on the tree, with sum–product as the cell) over a family of kernels. Identification is what makes that consistent, and for a finite-dimensional family of maps g_i the usual $n^{-1/2}$ rate applies. Boundedness of the φ_i is used twice: for Lemma 32, and for the compactness that produces exposed parents. The unbounded linear case is the LG tree of §7, where second moments do the same job.

8.6 V1: verifying Theorem 36 with the pure learner

Setting (fixed before the run). $d = 1$, $s = 3$, noise $\Psi = \psi^2$ with $\psi = 0.35$, and mean maps $\varphi_i(z) = \alpha_i + \beta_i\sigma(z) + \gamma_i \tanh(3\sigma(z))$ with random coefficients ($|\gamma_i| < |\beta_i|/3$, so each φ_i is strictly monotone and bounded). The root is drawn from the stationary law of the tree. For $d = 1$ the exposed parents are the two points at infinity, so condition (A) holds, and $\operatorname{Aut}(K)$ is generically trivial.

Learner. Maximum likelihood by EM, with no labels. Beliefs are represented exactly on a 161-point grid over $[-4, 4]$, and the cell is sum–product. The E-step is one upward and one downward belief-propagation pass. The M-step is a weighted least-squares fit of each φ_i in a 5-function basis $\{1, \sigma, \sigma^2, \sigma^3, \tanh 3\sigma\}$ (which contains the truth), plus a closed-form update of ψ. Its sufficient statistics are sums over grids of grid-local expected terms (the Fisher identity). The kernel is tied, so the statistics of every level are pooled. The root prior is the stationary law of the current kernel (stationarity, as in §7). Among 6 random restarts, the one with the highest likelihood is kept.

Predictions.

(P1) *Identification.* From unlabelled trees alone, the learned φ_i and ψ converge to the truth in leaf coordinates: no monotone reparametrization, no reflection. The root posterior mean reaches the R^2 of the Bayes oracle (belief propagation with the true kernel).

(P2) *Depth adds data.* The kernel error falls like $N_1^{-1/2}$ in the total number of bottom grids N_1 (parametric family, Corollary 27). At fixed N_1 it does not depend on how the grids are split into trees and depths. So a single tree suffices once it is deep enough.

Results. Two random true kernels, 4 data seeds each (`verify/ngt.py`, `verify/exp_ngt.py`; about 25 minutes on a laptop CPU). A first run had a normalization bug in the E-step, which weighted each grid's pair statistics by its message maximum instead of 1. It was found with a brute-force check and fixed. The numbers below are from the corrected learner. They differ from the first run by about 0.01, and no conclusion changed. Kernel error is the RMS of $\hat{\varphi}_i - \varphi_i$ under the true stationary law, averaged over i, in units of ψ.

Table 8: (P1) One unlabelled tree of depth L ($N_1 = 3^{L-1}$ bottom grids). Kernel error, relative error of ψ, and root R^2 of the learner against the Bayes oracle on 400 test trees. Means over 4 seeds.

		kernel 0				kernel 1			
L	N_1	error	ψ err	R^2	oracle	error	ψ err	R^2	oracle
5	81	0.394	0.073	0.680	0.711	0.463	0.143	0.185	0.267
6	243	0.221	0.022	0.710	0.711	0.251	0.069	0.222	0.240
7	729	0.132	0.005	0.711	0.713	0.158	0.014	0.173	0.179
8	2187	0.072	0.005	**0.711**	**0.711**	0.132	0.015	**0.132**	**0.137**

Table 9: (P2) Kernel error at fixed N_1, for different splits into U trees of depth L. Means over 4 seeds.

$N_1 = 243$	$(1, 6)$	$(3, 5)$	$(9, 4)$	$(27, 3)$	
kernel 0 / 1	0.22 / 0.25	0.25 / 0.32	0.22 / 0.41	0.18 / 0.36	
$N_1 = 729$	$(1, 7)$	$(3, 6)$	$(9, 5)$	$(27, 4)$	$(81, 3)$
kernel 0 / 1	0.13 / 0.16	0.16 / 0.19	0.10 / 0.22	0.12 / 0.19	0.11 / 0.21

Reading the results. (i) *P1 confirmed.* From one unlabelled tree, the learned kernel converges to the truth in leaf coordinates, and the root predictor reaches the Bayes oracle: 0.711 against 0.711 at $L = 8$ for kernel 0, and 0.132 against 0.137 for kernel 1. For kernel 0 the error falls by 0.55–0.60 per tripling of N_1, against the parametric $1/\sqrt{3} = 0.577$. Kernel 1 is weakly reconstructible (its oracle R^2 decays with depth), yet its kernel is still learned: the kernel comes from the grids, not from the root. (ii) *P2 confirmed for kernel 0 and roughly for kernel 1.* At fixed $N_1 \geq 243$, kernel 0's error shows no trend across splits. For kernel 1 the single deep tree is the best packaging, and shallow splits are 1.2–1.6× worse. That is a mild deviation from exact packaging-invariance, in the direction that favours depth. We have not explained it. (iii) *The one failure mode is the predicted ambiguity.* At $N_1 = 81$, some fits of kernel 0 land far from the truth (error $\approx 3\psi$). They all have the form $\hat{\varphi}_i(z) \approx \varphi_i(-z)$: the latents are reflected at odd heights and not at even ones. This is the isometry $T(x) = -x$, the only non-affine-shift candidate in $\mathrm{Aut}(K)$ for $d = 1$. It is *nearly* an automorphism here because kernel 0's maps are nearly odd (offsets $|\alpha_i| \leq 0.3$). The signature

matches: when the root height is even, the reflected fits still predict the root ($R^2 \approx 0.68$); when it is odd, they predict its negative ($R^2 < 0$). With 81 grids the likelihood cannot separate the two (the reflected fit even scores higher on its sample). From $N_1 = 243$ on, all seeds of both kernels resolve it. Theorem 36 and Proposition 29 predict exactly this: identification holds up to the isometries commuting with the kernel, and a near-commuting isometry costs data rather than identifiability. (iv) Very small samples ($N_1 \le 27$, i.e. $L \le 4$ for one tree, against 16 parameters) are not reported. Those fits are ill-posed, and some hit numerical underflow, since fixed.

8.7 V1′: a belief representation that scales

The grid of V1 is exact but costs (grid size)^d. We need a representation that scales and keeps the V1 guarantees.

Remark 39 (projected continuous beliefs are ill-posed for bounded kernels)**.** If the φ_i are bounded, the upward message $\beta_w(z) \propto \prod_i N(\varphi_i(z); \mu_i, v_i)$ tends to a positive constant as $\|z\| \to \infty$. It is therefore not normalizable, and projecting it onto a Gaussian (assumed-density filtering) is undefined. Projection has to be taken relative to a reference law, such as the stationary law π, by representing $\pi\beta_w$. In high dimension π is itself intractable, which is a second approximation.

Definition 40 (tied codebook tree)**.** Latents are symbols $a \in \{1, \dots, v\}$, and each symbol carries a value $c_a \in \mathbb{R}^d$ in leaf coordinates. A node with symbol a draws its children's symbols independently, $a'_i \sim M_i(\cdot \mid a)$, with the same M_i at every height. A leaf is the value of its own symbol plus noise: $x = c_{a'} + \tau\varepsilon$.

Beliefs over v symbols are always proper, and sum–product is exact at cost $O(s\,v^2)$ per cell. EM is exact: M_i is updated from pooled pair counts, and c and τ by weighted means. All statistics are grid-local. Symbols are points of leaf space, so "latents are the same kind of object as leaves" holds literally. The value of the root is predicted as $\mathbb{E}[c_{a_\epsilon} \mid X]$, with no labels: naming is carried by the codebook, and tying the M_i across the leaf level and all latent levels is what pins it (srg.pdf, Thm. 5.3(2)). The V1 grid model is the special case with a fixed codebook of 161 points and parametric M_i.

Predictions (fixed before the run), on the V1 data.

(Q1) With enough data, the codebook tree's root R^2 approaches the oracle's as v grows. The remaining gap is quantization error, which shrinks with v.

(Q2) Its data requirement is governed by its number of parameters, $\approx s\,v^2$, through the total number of bottom grids. So larger v needs more grids, and at small N_1 a small codebook wins (bias against variance). Depth supplies those grids as before.

(Q3) It is trained on trees of depth ≤ 8 and tested on depth-8 trees. Because the kernel is tied, a model trained on shallower trees transfers to deeper ones with no loss beyond its estimation error.

Results (`verify/codebook.py`, `verify/exp_codebook.py`, run on Colab; each EM from random initialization, best of 4 restarts by likelihood, 2 seeds; zero labels).

Reading Table 10. (Q2) *Confirmed.* The best codebook size grows with the data: $v = 4$–8 at $N_1 = 729$, $v = 8$ at 2187–6561, and $v = 16$ at 19683. A codebook too large for its data falls below zero R^2. The data requirement is set by the parameter count $\approx s\,v^2$, as predicted. (Q3) *Confirmed.* At fixed $N_1 = 6561$, models trained on depth-4 or depth-5 trees and tested on depth-8 trees come

Table 10: V1′: root R^2 of the tied codebook tree on depth-8 test trees, by codebook size v and training data (N_1 bottom grids in U trees of depth L). Oracle: 0.711 (kernel 0) and 0.137 (kernel 1).

		kernel 0				kernel 1			
N_1	(U, L)	$v = 4$	8	16	32	4	8	16	32
729	$(1, 7)$	0.59	0.59	0.34	0.15	0.12	−0.02	−0.31	−0.61
2187	$(1, 8)$	0.55	**0.63**	0.56	0.37	0.12	0.07	−0.10	−0.33
6561	$(3, 8)$	0.56	**0.67**	0.65	0.55	0.12	0.09	0.08	−0.01
6561	$(81, 5)$	0.54	**0.66**	0.62	0.52	0.13	0.09	0.09	−0.06
6561	$(243, 4)$	0.53	**0.64**	0.62	0.55	0.13	0.11	0.09	−0.03
19683	$(9, 8)$	0.55	0.66	**0.69**	0.61	0.13	0.10	0.11	0.12

within 0.01–0.03 of models trained on depth-8 trees. The tied kernel transfers to depths never seen in training. (Q1) *Partly confirmed.* With the most data, the codebook tree reaches 0.69 against the oracle's 0.71 at $v = 16$ (kernel 0). For kernel 1, where there is little signal to find, $v = 4$ is already within 0.01 of the oracle. But $v = 32$ still trails at $N_1 = 19683$ (0.61), and this run does not tell whether that is lack of data or EM local optima from random initialization. The parametric V1 learner reaches the oracle from $N_1 = 2187$. A free $v \times v$ table pays for its generality in data (about $10\times$ here). That argues for structured kernels (smooth or low-rank M_i) as the next refinement.

Follow-up: optimization, and a floor. We reran with coarse-to-fine EM ($v = 4 \to 8 \to 16 \to 32 \to 64$, splitting every symbol in two) on a GPU, adding $N_1 = 59049$ (`verify/codebook_torch.py`, `verify/exp_codebook_c2f.py`). At $N_1 = 19683$, coarse-to-fine lifts $v = 32$ from 0.61 to 0.67, so part of the shortfall was optimization. But at $N_1 = 59049$ kernel 0's R^2 is flat at 0.676–0.678 for *every* $v \geq 8$, about 0.035 below the oracle. Kernel 1 shows the same (about 0.11–0.13 against 0.137). So Q1 fails as stated: the gap does not close with v.

Diagnosis: a level mismatch in Definition 40. In the true process every child, leaf or latent, is its parent's map plus the same noise. In the codebook tree, latent children are exact symbols drawn by M_i, and only leaves carry the extra emission noise τ. Maximum likelihood lets τ absorb part of the leaf noise; the learned $\tau = 0.24$–0.28 is more than half the true noise variance, and it barely shrinks with v. So the tied M_i carries too little noise for the latent levels, their beliefs are over-confident, and the result is a bias floor that neither v nor data removes. This is a violation of "same type" that the definition let in.

Prediction Q4 (fixed before the run). Remove the emission noise. Each leaf's symbol is its quantile cell (the cells at $2v$ refine those at v, so coarse-to-fine splitting stays exact), and a symbol's value c_a is its cell mean. All noise then lives in the tied M_i, at every level, as in the true process. The model is consistent as $v \to \infty$. So the floor should disappear, and R^2 should approach the oracle as v and N_1 grow together.

Q4 confirmed. Removing the emission noise removes the floor. Kernel 0 reaches the oracle (0.702–0.712 against 0.711) at every codebook size and every N_1, where the noisy-emission tree was stuck at 0.68. For kernel 1 the best R^2 climbs toward the oracle as data grow (0.118, 0.122, 0.131 against 0.137), and the best v grows with N_1 (bias against variance, Q2 again). The lesson is a design rule taken from the theory, not a tuning trick. *Leaves must be generated exactly as latent children are*, so that one tied kernel carries all the stochasticity at every level. Any per-level extra noise breaks the tying and costs a floor. With that rule, the codebook tree is an exact, scalable belief representation that reaches the Bayes oracle from unlabelled trees.

Table 11: Q4: hard-quantized codebook tree (coarse-to-fine EM, GPU; `verify/exp_codebook_q.py`), root R^2 on depth-8 test trees, mean of 2 seeds, zero labels. Oracle 0.711 (kernel 0), 0.137 (kernel 1).

	N_1	$v = 4$	8	16	32	64	128
kernel 0	6561	0.706	0.709	0.707	0.707	0.702	
	19683	0.705	0.709	0.709	0.710	**0.712**	
	59049	0.705	0.708	0.708	0.709	0.709	0.708
kernel 1	6561	0.110	**0.118**	0.100	0.073	0.021	
	19683	0.116	**0.122**	**0.122**	0.120	0.104	
	59049	0.113	0.125	**0.131**	0.130	0.122	0.082

8.8 V2: more than one dimension

Setting (fixed before the run). $d = 2$, $s = 3$, $\psi = 0.35$, and $\varphi_i(z) = b_i + W_i\sigma(z) + U_i \tanh(3\sigma(z))$, with σ applied coordinatewise. W_i is a random rotation times singular values in $[1.3, 2.0]$, $\|U_i\| \le 0.3\,\sigma_{\min}(W_i)$, and $b_i \in [-0.3, 0.3]^2$. (A first choice, singular values in $[0.6, 1.2]$, left the root unrecoverable: oracle $R^2 = 0.02$–0.18. That made R1 untestable, so the process was strengthened before any learner was run on it; with $[1.3, 2.0]$ the oracle R^2 is 0.65–0.78.) The oracle is belief propagation with the true kernel on a 101×101 grid over $[-5, 5]^2$. It was validated against the exact one-dimensional oracle on a separable kernel (agreement to 0.006). Its Gaussian transition is applied as a separable blur followed by interpolation, since a dense matrix would be too large.

Learner. The hard-quantized codebook tree, unchanged except for the leaf quantizer. Quantile cells do not exist for $d > 1$, and the design rule of Q4 asks for deterministic cells without emission noise. Coarse-to-fine splitting needs *nested* cells. So the leaves are quantized by tree-structured vector quantization: each cell is split in two by 2-means, started along its principal axis. Cells at size $2v$ refine those at size v, a symbol's value is its cell mean, and EM, the root readout and zero labels are as in Q4.

Predictions.

(R1) No floor. The best root R^2 approaches the oracle's as v and N_1 grow.

(R2) The price of dimension is paid through v, not through depth. In $d = 1$, $v = 4$–8 came within 0.01 of the oracle. In $d = 2$ this needs a larger v, and so, through the parameter count $\approx s\,v^2$, more bottom grids. Tree depth still supplies those grids.

(R3) Packaging. At fixed N_1, 81 trees of depth 5 and 3 trees of depth 8 give about the same R^2 on depth-6 test trees.

Results (`verify/ngt_nd.py`, `verify/codebook_vq.py`, `verify/exp_v2.py`; Colab L4 GPU, about one minute per coarse-to-fine ladder; 2 seeds; zero labels).

Reading Table 12. (R1) *Confirmed.* No floor. The gap to the oracle closes as data grow: 0.019, 0.010, 0.007 for kernel 200, and 0.017, 0.010, 0.005 for kernel 201, at $N_1 = 6561$, 19683, 59049. (R2) *Confirmed.* In $d = 1$, $v = 4$–8 came within 0.01 of the oracle. In $d = 2$ this takes $v = 16$–32: about the square, as one would expect when resolving the latent to the same precision per axis. The data needed rise with it. $v = 32$ overtakes $v = 16$ only from $N_1 \approx 2 \times 10^4$, and over-large codebooks fall away at small N_1. Dimension is paid for in v, and depth still supplies the grids. (R3) *Confirmed.* At $N_1 = 6561$, 81 trees of depth 5 match 3 trees of depth 8 (0.511 against 0.521, and 0.727 against

Table 12: V2, $d = 2$: root R^2 of the hard-quantized codebook tree (tree-structured VQ leaves) on depth-6 test trees, by codebook size v and training data. Oracle 0.540 (kernel 200) and 0.744 (kernel 201).

	N_1 (U, L)	$v = 4$	8	16	32	64	128	256
k. 200	6561 $(3, 8)$	0.435	0.476	**0.521**	0.503	0.433	0.307	0.102
	6561 $(81, 5)$	0.436	0.468	**0.511**	0.502	0.427	0.275	0.123
	19683 $(9, 8)$	0.441	0.468	0.527	**0.530**	0.507	0.453	0.361
	59049 $(27, 8)$	0.437	0.471	0.523	**0.533**	0.530	0.512	0.464
k. 201	6561 $(3, 8)$	0.623	0.665	**0.719**	0.707	0.688	0.615	0.467
	6561 $(81, 5)$	0.287	0.669	**0.727**	0.722	0.687	0.618	0.503
	19683 $(9, 8)$	0.621	0.666	0.732	**0.734**	0.729	0.705	0.649
	59049 $(27, 8)$	0.621	0.661	0.732	**0.739**	0.737	0.735	0.712

0.719, at the best v). One seed at $v = 4$ for kernel 201 stopped in a poor local optimum, which pulls that single cell's mean down to 0.287. *Consequence for the architecture.* The pure learner carries over to several dimensions unchanged: exact sum–product on symbol beliefs, EM with grid-local statistics, zero labels, and nested vector quantization of the leaves. It reaches the Bayes oracle. Its cost in d dimensions is the codebook size, which grows roughly like (resolution)^d. That is the known price of discrete beliefs, and the reason structured (factorized or low-rank) kernels are the next step before high-dimensional data.

8.9 Head to head: the same architecture, two learning rules

The codebook tree is differentiable end to end in its kernel M_i (softmax parameters) and its values c. The leaf quantizer is a fixed preprocessing step shared by every rule. Autodiff of the log-likelihood through the upward pass and the stationary prior reproduces the grid-local gradient (Fisher identity) to 10^{-5}, against gradients of size 175. So backpropagation *on the likelihood* is our rule computed the long way round. The genuine alternative is backpropagation of a *root-label* loss: squared error of $\mathbb{E}[c_{a_\epsilon} \mid X]$ against the true root, pushed down through every level. The comparison below changes only the learning rule. Architecture, parameters, quantizer, initialization and test trees are identical.

Prediction (fixed before the run). First, where is the root information? The best *additive* predictor $\sum_\ell (a_\ell x_\ell + b_\ell x_\ell^2)$, fitted on 20k labelled trees, reaches 0.97–0.99 of the oracle's R^2 at depths 3–6 for both kernels (`verify/check_ngt_additive.py`; at depth 7 the 4374-feature fit over-fits). So this process is Gaussian-like in the sense of Proposition 43: the linear phase of backpropagation sees almost all of the signal. Hence:

(H1) Backprop's label requirement does *not* grow with depth here. The exponential cost of §9 needs root information held in interactions, and this process has almost none.

(H2) The gap is the one claim 1 predicts. The local rule uses no labels and needs fewer trees as depth grows (one tree at depth 8, Table 11). Backprop needs a roughly depth-independent number of labelled trees, set by its parameter count and label noise.

Backprop is given its best case: its learning rate is chosen per setting on a separate labelled validation set of 300 trees, which is not counted in its label budget.

Positive control. With 2048 labelled trees of depth 4, backprop through this architecture reaches $R^2 = 0.74$ (oracle 0.77 on that test set). The training code works, and any shortfall below reflects

labels, not a defect.

Table 13: Same architecture ($v = 8$ hard-quantized codebook tree), same P trees of depth L, same test trees. Root R^2, mean of 2 seeds. *Local*: EM, labels discarded. *Backprop*: root-label loss, P labels, best of 3 learning rates on a separate validation set (`verify/exp_h2h_ngt.py`, Colab GPU). Oracle 0.716, 0.711, 0.711.

L	rule	$P = 2$	8	32	128	512
4	local (0 labels)	0.08	0.56	0.67	0.70	**0.71**
	backprop (P labels)	−0.43	0.09	−0.18	0.57	0.66
6	local (0 labels)	0.67	**0.70**	0.71	0.71	0.71
	backprop (P labels)	−1.09	0.02	0.29	0.29	0.68
8	local (0 labels)	**0.70**	0.71	0.71	0.71	0.71
	backprop (P labels)	0.28	0.06	0.38	0.54	0.66

Reading Table 13. (H1) *Confirmed.* Backprop's label requirement does not grow with depth: with 512 labels it reaches 0.66, 0.68, 0.66 at depths 4, 6, 8. Its curve is noisy at small P, often worse than predicting the mean, even with learning-rate selection on held-out labels. This is the local-optimum behaviour already seen for backprop on the exact tied belief-propagation architecture (srg.pdf, §5). (H2) *Confirmed.* The local rule needs fewer trees the deeper they are: about 32–128 trees for $R^2 \geq 0.67$ at depth 4, 2–8 at depth 6, and 2 at depth 8, with zero labels throughout. Backprop needs about 512 labelled trees for 0.66 at every depth, and never reaches the oracle within that budget. At depth 8: 2 unlabelled trees give the local rule 0.70, while 512 labelled trees give backprop 0.66. *What this isolates.* The architecture, parameters, quantizer, initialization and test trees are identical; only the learning rule differs. Nor is the advantage a question of what the rule can express: backprop on the likelihood computes the same gradient as the local rule. The advantage is what each rule uses. Backprop from root labels learns from one number per tree, delivered through the root. The local rule learns from every grid of every tree, and depth multiplies the grids. On this Gaussian-like process the gap is exactly claim 1's, as predicted; the additional exponential backprop cost of §9 would require root information held in interactions.

8.10 Structured kernels

The free table $M_i \in \mathbb{R}^{v \times v}$ has two separate costs. (a) Its $s\,v^2$ parameters make the data requirement grow like v^2; that is why over-large codebooks collapse at small N_1 (Tables 11 and 12). (b) Beliefs have size v, and v must resolve the latent space, so v grows like $(\text{resolution})^{d_{\text{int}}}$. Here d_{int} is the intrinsic dimension of the data: tree-structured quantization places cells only where the data are, so the ambient dimension does not enter. A structure must remove (a) and must keep what the theory needs: exact sum–product, grid-local EM, and naming.

- *Low rank*, $M_i = A_i B_i$: exact, and it cuts compute. But its parameters still grow with v, and nothing ties it to leaf geometry, which is what fixes the naming.
- *Factorized over coordinate groups*: the child groups are coupled through the parent, so exact beliefs keep full size. The factorization buys nothing without giving up exactness.

- *Smooth kernel on the codebook*: the true model class, written on the adaptive codebook,

$$\log M_i[a' \mid a] = -\frac{\|c_{a'} - \varphi_i(c_a;\theta)\|^2}{2\psi^2} + \log \omega_{a'} - \log Z_i(a).$$

  Here φ_i is a small network in leaf coordinates, tied across heights, and $\omega_{a'}$ corrects for cell size. Sum–product stays exact. The parameters (θ, ψ) do not depend on v; only the v cell weights do. Naming is Theorem 36's setting, since the kernel lives in leaf coordinates. EM becomes a *generalized* EM. The E-step is unchanged, and the M-step raises $\sum_{i,a,a'} \mathrm{pair}_i(a, a') \log M_i[a' \mid a]$ by gradient steps on (θ, ψ, ω): gradients of the pooled grid-local statistics, still maximum likelihood. Refining the codebook reuses θ as it is.

Setting and predictions (fixed before the run). The same $d = 2$ processes, oracles and test trees as V2. $\varphi_i(z) = b_i + L_i z + B_i \tanh(A_i z + a_i)$ with 16 hidden units per child position: a generic map that does not contain the true σ-based form. Zero labels.

(S1) Data efficiency. The structured kernel comes within 0.01 of the oracle from a single depth-8 tree ($N_1 = 2187$). The free table needed $N_1 \approx 6 \times 10^4$.

(S2) No collapse at large v. Since the parameters do not grow with v, R^2 does not fall as v grows at fixed N_1. It rises or saturates, because v now sets only resolution.

(S3) Identification. The learned φ_i approach the true maps in leaf coordinates (error under the stationary law, in units of ψ) as N_1 grows, with no labels (Theorem 36).

First run: a selection flaw, diagnosed. In a first run (`verify/exp_structured.py`; 2 random restarts, the better one chosen by likelihood at $v = 8$ and then refined), one data seed of each kernel often failed, even at $N_1 = 19683$ ($R^2 \approx 0.2$, map error $\approx 5\psi$). The other seed confirmed S1–S3. A diagnostic ran 8 independent restarts per failing configuration, each carried through the whole ladder (`verify/exp_structured_diag.py`). (i) For $N_1 \geq 2187$ the restart with the highest *final* likelihood is always the good one (e.g. $R^2 = 0.744$ and map error 0.59ψ for kernel 201 at $N_1 = 6561$). The likelihood identifies the truth, as Theorem 36 says; the failures were EM local optima. (ii) Choosing at $v = 8$ was the flaw. At that coarse resolution the likelihood cannot separate the truth from *near-automorphism* solutions. These are the true maps conjugated by an isometry, the two-dimensional form of V1's reflection. They predict the root well ($R^2 \approx 0.74$), because the transformation alternates between heights and the root is at an even height. But their maps are far off ($\approx 5\psi$). Only at the finest codebook does the likelihood rank them below the truth. (iii) At $N_1 = 729$ even the finest likelihood prefers a near-automorphism solution. That is the data-limited ambiguity V1 showed at $N_1 = 81$. The corrected procedure is still maximum likelihood: 8 restarts, each carried through the full ladder, the final likelihood choosing. *Prediction for the corrected run:* the seed failures disappear for $N_1 \geq 2187$, and S1–S3 hold for every seed; at $N_1 = 729$ a near-automorphism solution may remain.

Reading Table 14. Every seed succeeds for $N_1 \geq 2187$, confirming the diagnosis. The single remaining failure is kernel 201, seed 1, at $N_1 = 729$: a near-automorphism solution (map error 4.7ψ) with $R^2 = 0.64$, as predicted. (S1) *Confirmed, with one qualification.* From one depth-8 tree ($N_1 = 2187$) kernel 200 comes within 0.007 of the oracle; kernel 201 is within 0.018 there and within 0.002 from $N_1 = 6561$. The free table needed $N_1 = 59049$ to reach 0.533 and 0.739: the structured kernel matches it with 9–27× fewer grids. (S2) *Confirmed.* R^2 no longer collapses at large v. For $N_1 \geq 2187$ it rises or saturates up to $v = 256$, since v now sets only resolution (per-seed

Table 14: Structured kernel, corrected selection (`verify/exp_structured_v2.py`, Colab GPU): root R^2 at $v = 256$ and map error (units of ψ), mean of 2 seeds, zero labels. Free table (V2, Table 12): best R^2 over v.

	kernel 200 (oracle 0.540)			kernel 201 (oracle 0.744)		
N_1	R^2	map err	free table	R^2	map err	free table
729	0.479	1.04	–	0.649	3.32	–
2187	**0.533**	0.60	–	0.726	1.23	–
6561	**0.540**	0.51	0.521	**0.742**	0.60	0.727
19683	**0.539**	0.50	0.530	**0.743**	0.57	0.734
59049			0.533			0.739

values in `exp_structured_v2_results.jsonl`). At $N_1 = 729$ a mild decline remains ($0.51 \to 0.48$ for kernel 200). (S3) *Confirmed.* The learned maps approach the true maps in leaf coordinates as N_1 grows, from 1–3ψ at $N_1 = 729$ to about 0.5ψ, with no labels. The remaining 0.5ψ is a floor: the network is evaluated at cell means rather than at the latent itself, and it approximates the true σ-based map without containing it. *Lesson for the procedure.* Restarts must be compared at the finest resolution. A coarse codebook cannot distinguish the truth from its near-automorphisms. It cannot, because those solutions differ only in detail the coarse cells average away.

8.11 What this settles, and what remains

Settled, for every tied tree process:

- the object that context identifies is the belief, i.e. the upward BP message, and this fixes the state space and the cell (Theorem 21, Remark 24);
- estimating it depends on the data only through the total number of bottom grids, so depth adds data and one deep example suffices (Theorem 26, Corollary 27);
- naming is identified up to $\mathrm{Aut}(K)$ (Theorem 28), which under Gaussian noise is a finite-dimensional group of commuting isometries (Proposition 29);
- for bounded nonlinear Gaussian trees, all of this follows from the laws of the bottom two levels alone: the noise, the kernel in gauge coordinates, and the naming up to $\mathrm{Aut}(K)$, with no grouping by parent and no labels (Theorem 36).

The discrete results of §3–4 and the linear results of §7 are special cases. *Check (soft regime, never tested in §3–4).* A generic soft kernel on $\mathcal{Z} = \{0, 1, 2\}$, $s = 3$, with one planted likelihood-proportional pair of tuples (`verify/check_general.py`). From one tree of depth 11, the empirical context laws match $\Lambda_n \beta_t^n$ to within pure sampling noise (mean L^1 error 0.267 against a noise-only baseline of 0.255). The planted pair has identical theoretical context laws, and its empirical distance (0.279) equals the noise-only expectation (0.294). With $n_2 = 729$ fixed and split into $1, 3, 9, 27, 81$ trees, the normalized error is flat (2.06–2.15), as Theorem 26 predicts.
Remaining.

1. *Beyond Gaussian noise.* Hypothesis (N) is now a theorem for bounded nonlinear Gaussian trees (Theorem 36). Its proof uses Gaussianity in three places: the analytic characteristic function (Lemma 32), deconvolution, and the Radon–Nikodym argument. Other noise laws, and statistical rates for the maximum-likelihood learner, remain open.

2. *Backprop.* Treated in §9.

3. *A testbed* that follows from the above: a nonlinear Gaussian tree with a single "folding" parameter, from nearly linear (LG) to strongly many-to-one (RHM-like). Theorems 21–28 predict zero-label recovery from one deep tree across the whole range, and the conjecture predicts a backprop gap that grows with folding.

9 Why learning from root labels slows with depth

The context learner uses one signal: the dependence between *siblings*. Its strength does not depend on depth, and a single example supplies s^{L-2} conditionally independent copies of it (Theorems 4 and 26). Backpropagation from root labels uses another: the dependence between the *root label* and whatever function of the leaves the network currently computes. This section shows that the second signal, for the functions a network can form early, decays geometrically with depth at a rate q_1 that is a property of the kernel. It then locates exactly when that decay produces an exponential label cost: never in Gaussian trees, and always in tied grammars whose root is determined by the leaves while $q_1 < 1$. For the grammars of Table 1, q_1 accounts for the depth dependence of the backprop labels measured in Table 3.

9.1 Local features and the linear transfer operator

Consider a tied tree process (Definition 19) and a node w at depth j (distance from the root) with marginal law π_w. For its path $\pi(w) = (i_1, \dots, i_j)$ from the root, let $K_{\pi(w)} : L_0^2(\pi_w) \to L_0^2(\pi_\epsilon)$ be $(K_{\pi(w)}b)(z) = \mathbb{E}[b(z_w) \mid z_\epsilon = z]$, where L_0^2 denotes centred functions. Put

$$\mathcal{A}_j = \sum_{|w|=j} K_{\pi(w)} K_{\pi(w)}^*, \qquad q_1 = \limsup_{j\to\infty} \|\mathcal{A}_j\|^{1/j}.$$

$\mathcal{A}_j$ obeys the recursion $\mathcal{A}_j = \sum_i K_i \mathcal{A}_{j-1}^{[i]} K_i^*$, where K_i is the single-edge operator to child i and $\mathcal{A}^{[i]}$ is the same object for the subtree of child i. Hence $\|\mathcal{A}_j\| \le (s\lambda^2)^j$ and $q_1 \le q = s\lambda^2$, with λ the largest maximal correlation of a parent–child edge. If all nodes have the same marginal, then $\mathcal{A}_j = \mathcal{T}^j(I)$ for the positive map $\mathcal{T}(X) = \sum_i K_i X K_i^*$, and q_1 is its spectral radius. q_1 can be much smaller than q. λ bounds each edge separately, while $\mathcal{A}_j$ composes *different* channels along each path, whose leading singular directions need not align. We call q_1 the *linear Kesten–Stigum number* of the kernel. For a single channel repeated at every position it is the classical $s\lambda_2^2$.

A *k-local feature* is $F = \sum_{w\in V^{(k)}} h_w(X_w)$, where X_w denotes the leaves below w and the h_w are arbitrary square-integrable functions. For $k = 0$ these are the *additive* (degree-one) features $\sum_\ell a_\ell(x_\ell)$ of the leaves. Let α be the *additive conditional-variance constant*: the infimum, over internal nodes u and functions $b_1, \dots, b_s$ with $\sum_i \operatorname{Var} b_i(z_{ui}) > 0$, of

$$\mathbb{E}\big[\operatorname{Var}\big(\textstyle\sum_i b_i(z_{ui}) \,\big|\, z_u\big)\big] \Big/ \textstyle\sum_i \operatorname{Var} b_i(z_{ui}).$$

It measures how far a sum over siblings is from being a function of the parent. For conditionally independent kernels $\alpha \ge 1 - \lambda^2$.

9.2 The label signal of local features

Theorem 41 (label signal decays at rate q_1)**.** *For every $g \in L_0^2(\pi_\epsilon)$, every $0 \le k < L$ and every k-local feature F,*

$$\operatorname{corr}\big(g(z_\epsilon), F\big)^2 \;\le\; \frac{\|\mathcal{A}_{L-k}\|}{\alpha}.$$

Hence the best correlation between the root and any k-local feature decays like $q_1^{(L-k)/2}$.

Proof. Let $b_w(z_w) = \mathbb{E}[h_w(X_w) \mid z_w]$ and $B = \sum_w b_w(z_w) = \mathbb{E}[F \mid Z_k]$ (Z_k are the latents at height k; by the Markov property $\mathbb{E}[h_w(X_w) \mid Z_k]$ depends on z_w only). *Signal.* X_w is conditionally independent of z_ϵ given z_w, so $\mathbb{E}[g(z_\epsilon)h_w(X_w)] = \langle g, K_{\pi(w)}b_w\rangle$. Since $\mathbb{E}g = 0$ and each $K_{\pi(w)}$ maps constants to constants, $\operatorname{Cov}(g, F) = \langle g, \sum_w K_{\pi(w)}\bar{b}_w\rangle$ with $\bar{b}_w = b_w - \mathbb{E}b_w$. The operator $(\bar{b}_w)_w \mapsto \sum_w K_{\pi(w)}\bar{b}_w$ from $\bigoplus_w L_0^2(\pi_w)$ has squared norm $\|\sum_w K_{\pi(w)}K^*_{\pi(w)}\| = \|\mathcal{A}_{L-k}\|$. Therefore $\operatorname{Cov}(g, F)^2 \le \|g\|^2\|\mathcal{A}_{L-k}\| \sum_w \operatorname{Var} b_w$. *Noise.* $\operatorname{Var} F \ge \operatorname{Var} B \ge \mathbb{E}\operatorname{Var}(B \mid Z_{k+1})$. Given Z_{k+1}, the sibling groups $(z_{u1}, \dots, z_{us})$, $u \in V^{(k+1)}$, are independent. So $\mathbb{E}\operatorname{Var}(B \mid Z_{k+1}) = \sum_u \mathbb{E}\operatorname{Var}(\sum_i b_{ui}(z_{ui}) \mid z_u) \ge \alpha \sum_w \operatorname{Var} b_w$. Divide. □

Three things are worth noting. The block size k enters only through $L - k$: a feature of any fixed size is attenuated by all the levels above it. Tying is not used. And the bound concerns correlation with the *root*, which is what a root label reveals. The dependence between siblings is untouched by it.

9.3 The first phase of backpropagation sees only this signal

Proposition 42 (the linear phase)**.** *Let F_θ be any network whose output and parameter derivatives are additive in the leaves. This includes every deep* linear *tree network, tied or untied, of any width, with a linear leaf encoder. Let the labels be $y = g(z_\epsilon) \in \{\pm 1\}$ with $\mathbb{E}g = 0$ (any balanced binary coding of the classes), and let $\rho^2 = \sup_{F \text{ additive}} \operatorname{corr}(g, F)^2 \le \|\mathcal{A}_L\|/\alpha$. For the square loss on P examples, the gradient is $\hat{G} = \frac{1}{P}\sum_\mu (F_\theta(x_\mu) - y_\mu)\nabla F_\theta(x_\mu)$. Its label-dependent part $\hat{G}^y = -\frac{1}{P}\sum_\mu y_\mu \nabla F_\theta(x_\mu)$ satisfies, coordinate by coordinate,*

$$\frac{|\mathbb{E}\hat{G}_j^y|}{\operatorname{sd}(\hat{G}_j^y)} \;\le\; \sqrt{P}\,\frac{\rho}{\sqrt{1-\rho^2}}.$$

Unless $P \gtrsim (1-\rho^2)/\rho^2 \approx \alpha/\|\mathcal{A}_L\|$, the labels enter every gradient step below their own sampling noise.

Proof. Write $D_j = \partial_j F_\theta$, an additive feature. Then $\mathbb{E}\hat{G}_j^y = -\operatorname{Cov}(g, D_j)$, and since $g^2 = 1$, $P \operatorname{Var}\hat{G}_j^y = \mathbb{E}D_j^2 - \operatorname{Cov}(g, D_j)^2 \ge (1-\rho^2)\operatorname{Var} D_j$, while $|\operatorname{Cov}(g, D_j)| \le \rho \operatorname{sd}(D_j)$. The remaining part $\frac{1}{P}\sum_\mu F_\theta \nabla F_\theta$ does not involve labels. □

For a smooth cell $f(c) = \sigma(\sum_i W_i c_i + \beta)$ with $\sigma(0) = 0$, $\sigma'(0) = 1$, the network at parameters $\varepsilon\theta$ is, to leading order in ε, a deep linear tree network. So Proposition 42 describes the first phase of gradient descent from small initialization. Interactions between leaves, which is where synonyms live, enter only at higher order.

9.4 Gaussian trees have no gap

Proposition 43. *In the LG tree of §7, tied or untied, and for a linear label $g(z_\epsilon) = a^\top z_\epsilon$, the Bayes predictor $\mathbb{E}[a^\top z_\epsilon \mid X]$ is an affine, hence additive, function of the leaves. So the best additive feature is the best feature of any kind: the degree-one label signal equals the Bayes R^2 at every depth. There is no regime in which the root is reconstructible while the additive label signal decays. Whenever the root can be predicted at all, the linear phase of backpropagation already sees the whole signal.*

Proof. Conditional expectations of jointly Gaussian vectors are affine, and $\operatorname{corr}(Y, F)^2 \le \operatorname{corr}(Y, \mathbb{E}[Y \mid X])^2$ for every $F = F(X)$. □

This explains the second half of §7: backprop's label cost did not grow with depth in the linear model (Table 7). The exponential cost is not a consequence of self-similarity or of depth as such. It requires information about the root that sits in *interactions* between leaves, which additive features cannot see.

9.5 Where the gap lives: tied grammars

Corollary 44 (separation at the start of learning)**.** *In the tied RHM with trivial* $\mathrm{Aut}(D)$ *and non-degenerate histograms:*

1. *the root is a deterministic function of the leaves at every depth (parsing), so the best feature of any kind has correlation* 1*;*
2. *the best additive feature has* $\mathrm{corr}^2 \le \|\mathcal{A}_L\|/\alpha$*, which is* $O\big((q_1+\varepsilon)^L\big)$*;*
3. *the context learner recovers the grammar with zero labels from* $O(s^{-(L-2)})$ *examples (Corollary 9);*
4. *the linear phase of backpropagation needs* $P \gtrsim \alpha\,\|\mathcal{A}_L\|^{-1}$ *labelled examples before any gradient coordinate carries label information above its noise (Proposition 42).*

When $q_1 < 1$*, the ratio between (4) and (3) grows like* $(s/q_1)^L$*.*

Table 15: The linear Kesten–Stigum number q_1 against the old constant $q = s\lambda^2$ for the grammars of Table 1. q_1 is the growth rate of $\|\mathcal{A}_j\|$ for $j = 7..12$, computed exactly with a uniform root; α is minimized over the node marginals up to depth 8. Code: `verify/check_claim2.py`.

grammar	0	1	2	3	4	5	6	7
$q = s\lambda^2$	1.28	1.16	1.88	1.63	1.68	1.32	0.94	1.57
q_1	0.40	0.41	0.93	0.49	0.46	0.45	0.37	0.50
α	0.084	0.049	0.035	0.070	0.091	0.146	0.169	0.097

Table 16: Grammar 0, the grammar of Table 3. $\|\mathcal{A}_L\|$ exactly; the best additive correlation estimated by Monte Carlo (held-out top canonical correlation between the root and one-hot leaves, 400k trees); and the backprop labels P^*_{bp} of Table 3.

L	$\|\mathcal{A}_L\|$	bound $\|\mathcal{A}_L\|/\alpha$	best additive corr^2 (MC)	P^*_{bp}	$P^*_{\mathrm{bp}}\|\mathcal{A}_L\|$
2	0.542	(6.45)	0.480	512	277
3	0.194	(2.31)	0.169	2048	397
4	0.088	1.05	0.074	4096	361
5	0.033	0.39	0.029	8192	270
6	0.013	0.15	–	16384	213

Reading Tables 15 and 16. (i) The old constant q exceeds 1 for seven of the eight grammars. Taken at face value, it would predict no growth of backprop's label cost with depth. q_1 is 0.37–0.50 for seven grammars. The channels to the two child positions are informative individually ($\lambda = 0.69$–0.97), but their products along a path lose signal quickly. (ii) The Monte Carlo correlation is 0.84–0.89 times $\|\mathcal{A}_L\|$ at every depth. So the *rate* q_1 is attained. The factor $1/\alpha$ in Theorem 41 is loose for

these grammars (and the bound is vacuous for $L \leq 3$), so the constant can be improved. (iii) Over $L = 2..6$, backprop's measured label requirement grows 32×, while $P^*_{\text{bp}}\|\mathcal{A}_L\|$ stays within 213–397. The predicted growth $1/q_1 \approx 2.3$–2.5 per level matches the measured ≈ 2.4. This agreement is a consistency check, not a proof. The data are one grammar and two seeds of standard (not small) initialization, and they extend beyond the linear phase that Proposition 42 covers. (iv) *A prediction not yet tested.* Grammar 2 has $q_1 \approx 0.93$. On it, backprop's label cost should barely grow with depth, while the context learner behaves as for the other grammars.

9.6 Experiment: grammar 0 against grammar 2

We tested prediction (iv) with the protocol of Table 3: the untied deep CNN, 3 seeds per point, $L = 2..7$, P up to 131072, on a Colab L4 GPU. The code is `verify/exp_claim2_bp.py`, and the raw results are `exp_claim2_bp_results.jsonl` and `exp_claim2_bp_long.jsonl`. The predictions were written down before the run. From $\|\mathcal{A}_L\|$, the label cost P^* should grow about 2.4× per level for grammar 0 and about 1.1× per level for grammar 2. We also compare the CNN with the best classifier that uses only additive features (least squares on one-hot leaves, $\geq$ 60k trees, `verify/check_claim2_additive_acc.py`).

Table 17: Backprop from root labels. P^* is the smallest P with mean test accuracy ≥ 0.9. "Before the jump" is the test accuracy at $P^*/2$ (at $P^*/4$ in brackets). "Additive ceiling" is the held-out accuracy of the best additive classifier. Chance is 0.125. †At $P = 131072$ (65536).

	grammar 0 ($q_1 \approx 0.40$)			grammar 2 ($q_1 \approx 0.93$)		
L	P^*	before the jump	additive ceiling	P^*	before the jump	additive ceiling
2	512	0.84 (0.57)	0.49	512	0.90 (0.67)	0.65
3	2048	0.77 (0.40)	0.32	1024	0.63 (0.45)	0.51
4	4096	0.44 (0.23)	0.23	4096	0.89 (0.48)	0.43
5	8192	0.24 (0.16)	0.18	4096	0.48 (0.32)	0.38
6	16384	0.14 (0.13)	0.15	8192	0.51 (0.29)	0.34
7	> 131072	0.13 (0.13)†	0.13	8192	0.28 (0.24)	0.30

What the experiment says. (i) *Confirmed: the phase before the jump is additive learning.* At every depth and for both grammars, the CNN's accuracy at $P^*/4$ lies within 0.08 of the additive ceiling, and full accuracy follows within a further factor of 4 in P. Where the jump is late ($L = 6, 7$), the accuracy approaches the ceiling from below: $0.23, 0.25, 0.29$ against 0.34, and $0.20, 0.23, 0.24, 0.28$ against 0.30. This is the linear phase of Proposition 42, and the ceiling it reaches is set by the additive features, as Theorem 41 bounds. (ii) *Refuted: q_1 does not set the full-accuracy label cost through $L = 6$.* Grammar 2 is not flat. Its P^* grows 16× from $L = 2$ to 6, against 32× for grammar 0: grammar 2 is easier by a roughly constant factor, not by a different rate. Nor does any single eigenvalue of $\mathcal{A}_L$ fix a constant shared by both grammars: grammar 2's P^* grows faster than even its smallest eigenvalue decays. The jump itself is governed by something other than additive signal. Because root information sits in interactions between leaves, we expect it to be governed by those interactions. (iii) *At $L = 7$ the grammars separate sharply.* Grammar 2 still learns at $P = 8192$. Grammar 0 stays at chance for every $P \leq 131072$, and still at chance with 5× the training (200 epochs at $P = 16384$ and 32768, 3 seeds each). Its predictions are essentially constant throughout. That is exactly the depth at which grammar 0's additive ceiling reaches chance (0.133). Where the linear phase has nothing to climb, gradient descent does not leave the trivial solution within our

budget. Whether many more labels or many more steps would eventually free it, we cannot say. Caveats: one architecture without normalization; 3 seeds; powers-of-two grids in P.

9.7 Head to head: one unlabelled tree against backprop

On the same two grammars and the *same* test sets, we ran the zero-label context learner of §§3–4. It is given P unlabelled trees and knows only (v, m, s), and it doubles P until all 3 data seeds recover the grammar exactly, i.e. test accuracy 1 (`verify/exp_head2head.py`). Below the threshold the learner is all-or-nothing: a wrong partition leaves most test trees unparsable.

Table 18: Examples needed. Context learner: unlabelled trees for exact recovery (3/3 seeds), with 0 labels. Backprop: labelled examples for test accuracy ≥ 0.9 (Table 17). Backprop was not run beyond $L = 7$.

	L	2	3	4	5	6	7	9	11
grammar 0	context, unlabelled	512	512	256	64	64	**32**	8	**1**
	backprop, labelled	512	2048	4096	8192	16384	$>$ **131072**	–	–
grammar 2	context, unlabelled	512	256	128	64	32	**16**	4	**1**
	backprop, labelled	512	1024	4096	4096	8192	**8192**	–	–

The two learners move in opposite directions with depth. The context learner needs about 1024–2048 bottom grids at every depth, however they are packaged, so the number of examples halves per level, down to a single unlabelled tree at $L = 11$ (claim 1). Backprop's labelled requirement grows with depth. At $L = 7$ it stands at 8192 for grammar 2, and above 131072 for grammar 0, where it never leaves chance. The two curves start equal at $L = 2$ (512 each). At $L = 7$ the ratio is 512 for grammar 2 and more than 4096 for grammar 0, labelled examples against unlabelled ones. Each context fit takes 1–3 s on a laptop CPU, against GPU minutes per backprop point. The context learner's dependence on the grammar is only through Δ and $\gamma_{\min}$ (Theorem 4), which are similar for the two grammars. Backprop's dependence on the grammar is large: q_1 and the additive ceiling decide whether it moves at all.

9.8 What remains

1. *Beyond the linear phase (revised by §9.6).* The experiment separates two things. The *additive phase* is described by the theory: its ceiling and its dependence on the kernel through $\mathcal{A}_L$. The *jump* to full accuracy grows about 2× per level in both grammars, so it is not controlled by q_1. The revised conjecture has two parts. (a) The jump's label cost is governed by the label correlation of the interaction (synonym) structure. That structure is similar across random grammars of the same (v, m, s), which is why P^* grows alike in both. (b) When the additive ceiling reaches chance, backprop from a trivial start has no signal to follow and can stall entirely, as grammar 0 did at $L = 7$. Both parts need theory for the nonlinear phase. The local rule is unaffected by either, because it never uses the label.

2. *Sharp constants.* Replace $1/\alpha$ by a quantity that makes Theorem 41 tight. Table 16 suggests the truth is close to $\|\mathcal{A}_L\|$ itself.

3. *A testbed, designed from the theory.* The gap needs three things at once: $q_1 < 1$ (label signal decays); distinct sibling contexts, $\Delta > 0$ (the local rule works); and a reconstructible root. Proposition 43 rules out Gaussian kernels. Nonlinear kernels whose position channels

are misaligned are candidates. The first experiment is the grammar 2 versus grammar 0 prediction above.

4. *Measuring q_1 on real data.* $\|\mathcal{A}_j\|$ is the root correlation of the best additive feature at depth j. Estimating how it decays with depth on a dataset would predict, before any training, whether backprop's label cost on that data will grow with depth.

10 Real images: the image tree on CIFAR-10

The structured kernel of §8.10 reaches the Bayes oracle from one unlabelled tree, but only on processes that *are* tied trees with a given latent. A real image comes with no latent, is not exactly a tree process, and its class is not the root of a tied process. Before any run on real data, the theory has to say three things: what the state of an image region is (§10.1), why a linear head on the root is the right readout and what it can reach (§10.2), and how much data depth supplies on images this small (§10.3). The predictions for a joint classification–generation experiment on CIFAR-10 follow from these (§10.4).

10.1 The state of an image region

Definition 45 (thumbnail pyramid)**.** Let $y \in [0,1]^{S\times S\times 3}$ with $S = 2^{L+1}$. The quadtree of height L has 4^{L-c} nodes at height c; a node at height c owns a square of side 2^{c+1}. Its *thumbnail* $t_w \in \mathbb{R}^{2\times 2\times 3}$ is the 2×2 block average of its square; at height 0 it is the 2×2 patch of pixels itself. Child $i \in \{0,1,2,3\}$ of w (top-left, top-right, bottom-left, bottom-right) has mean colour $t_w[i]$, and its *detail* $\delta_{wi} \in \mathbb{R}^9$ is the orthonormal Haar transform of t_{wi} without its mean (horizontal, vertical and diagonal differences, three channels).

The thumbnail of a parent is the average of its children: the value part of the state is *parent-determined* (Corollary 23(1)). Block averaging is the gauge in which Stage 1 measured the parent→detail rule to be the same across a 32× range of scales, so it is the natural naming of the latent in leaf coordinates.

Lemma 46 (exact change of variables)**.** *The map $y \mapsto \big(t_{\text{root}}, (\delta_w)_{w\neq\text{root}}\big)$ is a linear bijection of $\mathbb{R}^{3S^2}$ with $\log|\det| = -3n\log 2$, where $n = \sum_{c<L} 4^{L-c}$ is the number of non-root nodes. Hence any law of the root thumbnail and the details is a density on images, and*

$$\log p(y) = \log p(t_{\text{root}}) + \sum_{w\neq\text{root}} \log p(\delta_w \mid \cdots) - 3n\log 2.$$

Proof. Go from the leaves up. For one child and one channel, the four values (a,b,c,d) map to $(2m,H,V,D)$ with m their mean; this is an orthogonal change of basis, so $(a,b,c,d)\mapsto(m,H,V,D)$ has determinant $1/2$. The means become the parent's thumbnail, and the argument repeats. There are $3n$ such factors. (Checked numerically, `verify/test_imtree.py`.) □

Consequence: no codebook. By Theorem 21(4), a parent-determined value has point beliefs. The codebook of Definition 40 existed to represent beliefs about a hidden continuous value. On the pyramid the continuous part of every state is observed at every height, so its belief is a point mass, and the structured kernel of §8.10 can be used in closed form: $\delta_{wi} \mid t_w \sim N(\varphi_i(t_w), \Psi)$. This is the same kernel, with the codebook value c_a replaced by the observed thumbnail, and with noise confined to the detail subspace, which is what the constraint "child mean = parent pixel" requires.

Proposition 47 (what must stay hidden)**.** *In a tree whose state is the thumbnail alone, sibling details are conditionally independent given the parent thumbnail. If, on the data, some statistic of sibling details is dependent given the parent thumbnail, then by Theorem 21(1) two regions with the same thumbnail can have different sibling-context laws, and a sufficient state must separate them. The minimal state is then* (t, κ)*, where* κ *is the context class (the belief about the parent beyond its thumbnail).* κ *is not a function of the region's thumbnail, and it is not parent-determined, so it stays hidden and its belief is a genuine distribution.*

Proof. The first sentence is the tree Markov property. The rest is Theorem 21(1)–(3) applied to the part of the state that the thumbnail does not determine. □

If κ takes H values, its beliefs are H-vectors and sum–product is exact at cost $O(H^2)$ per edge. This gives the model.

Definition 48 (image tree IT(H))**.** The state of a node is (t, h), with t its thumbnail and $h \in \{1, \dots, H\}$ a hidden context type. With the same kernel at every height, child i of a node in state (t, h) draws

$$h'_i \sim T_i[h, \cdot\,], \qquad \delta_i \mid t, h'_i \sim N\big(A_{i,h'_i}\, g_i(t) + b_{i,h'_i},\ \Sigma_{i,h'_i}\big), \qquad g_i(t) = \tanh(W_i t + c_i),$$

and its thumbnail is $t_i = t[i]\mathbf{1} +$ (inverse Haar of δ_i). The root draws $h_R \sim \pi$ and $t_R \mid h_R \sim N(m_{h_R}, S_{h_R})$. Leaves (height 0) are generated exactly like every other child (the design rule of Q4). g_i is the nonlinear structured part: a small network in leaf coordinates, tied across heights. IT(1) is the structured kernel of §8.10 on images with nothing hidden. In the *untied* variant every parent height has its own copy of (T, A, b, Σ, W, c).

Remark 49 (heavy tails come from the types)**.** Given the parent, a child's detail is a mixture over h' of Gaussians with different covariances: a scale mixture. Stage 1 found details with kurtosis 7–13 at every scale. IT(1) can produce excess kurtosis only through the variation of $\varphi_i(t)$ across parents; IT(H) also produces it through the type. The hidden part of the state is where the non-Gaussian structure of images must live.

Learning is maximum likelihood by generalized EM, as in §8.10, with zero labels. The E-step is exact sum–product over types. The M-step takes gradient steps on the expected complete-data log-likelihood, a sum of grid-local terms; by the Fisher identity its gradient equals that of the marginal log-likelihood (checked to 10^{-9}, `verify/test_imtree.py`, together with brute-force enumeration of all type configurations). Images are dequantized with uniform noise, $y = (x + u)/256$, so that Lemma 46 gives an upper bound on bits per dimension.

10.2 Readout: the linear head on the root is the Bayes rule

Proposition 50 (root readout)**.** *Add the label* y *as one more child of the root, with kernel* $\ell(y \mid z_R)$ *(a* label leaf*). Then for every image* X

$$P(y \mid X) = \sum_z \beta_R(z)\, \ell(y \mid z),$$

where β_R *is the root belief (the normalized upward message). The Bayes classifier is* linear in the root belief*. For* IT(H) *with* ℓ *depending on the type only,* $P(y \mid X) = \sum_h \beta_R(h) \ell(y \mid h)$*.*

Proof. The label leaf is a child of the root, so y is independent of X given z_R. Condition on z_R. □

The label kernel ℓ is learned by the same EM, with the label leaf observed on labelled images and marginalized on the others. It is the only thing labels touch, $H(C-1)$ numbers for C classes, independent of depth and image size. That labels are needed there, and only there, is Theorem 11 at the root: the root has no context, so no amount of unlabelled data names its types as classes.

Corollary 51 (ceiling of the root)**.** *No head on the root state beats the Bayes accuracy of y from (t_R, β_R) (data processing). For* IT(1)*, β_R is trivial and the ceiling is the Bayes accuracy from the 2×2 thumbnail of the whole image.*

What the theory does *not* say is that likelihood-trained root types line up with classes. The root belief is the image's evidence about the root state of a model trained for pixel likelihood; how much of the class it carries is an empirical property of images, which the experiment measures.

10.3 Depth adds data: how much, on images this small

A CIFAR image is a tree of height $L = 4$ with 340 child edges. Theorem 26 says only the total number of grids enters. For a parametric kernel the amount is set by the classical accounting.

Proposition 52 (excess risk, tied against untied)**.** *Let the maximum-likelihood estimate of a regular model with k parameters be fitted on N i.i.d. images. Its expected excess test negative log-likelihood per image is $k^*/(2N) + o(1/N)$, with $k^* = \mathrm{tr}(J^{-1}I)$ ($= k$ if the model is well specified). Hence:*

1. *the tied model with k kernel parameters has excess $\approx k/(2N)$ per image, spread over all heights in proportion to their numbers of detail coordinates, i.e. $\approx k/(2N \cdot 3S^2)$ per coordinate at* every *height;*
2. *the untied model has $\approx k$ parameters per height and total excess $\approx Lk/(2N)$ (each height pays for its own copy once: Lemma 12, depth–frequency cancellation, for the total likelihood);*
3. *but at height c its excess per detail coordinate is $\approx k/(2N \cdot 9 \cdot 4^{L-c})$. At the top height (4 children) this is $3S^2/36 \approx 85$ times the tied model's, for CIFAR.*

Proof. The first statement is the standard expansion of the MLE under regularity (Takeuchi). (1) If the edges of all heights are statistically alike, each coordinate carries the same share of the Fisher information, so the excess is spread evenly per coordinate. (2)–(3) The untied parameters at height c are estimated only from the $9 \cdot 4^{L-c}$ coordinates per image of that height, and their excess falls entirely on those coordinates. □

So on total likelihood tying buys a factor $L = 4$, and on the top scale, which is what the root belief and the global layout of samples depend on, a factor ≈ 85. *One image is not enough* on CIFAR: one image has 3072 coordinates and IT(16) has 26,032 parameters. What depth buys is that $N \approx$ a few hundred images reach the regime where the tied kernel is within 0.02 bits/dim of its limit, and that the top of the tree is learned as well as the bottom.

10.4 Predictions (fixed before the run)

Models: IT(H) with g_i of 32 hidden units; $H \in \{1, 16, 128\}$ tied (3,127, 26,032 and 253,952 parameters) and $H = 16$ untied (99,760). All are trained with zero labels. Heads use the root state only.

Calibration of the data (`verify/cifar_precheck.py`, run before any learner; Table 19). Given the parent thumbnail (an MLP fitted per child position), the residual log detail energies of siblings

are correlated at 0.22–0.35 at every height, so Proposition 47 applies at every scale. Horizontally adjacent *cousins*, which meet across a parent boundary, are correlated even more (0.39–0.45): a tree can carry that dependence only through their common ancestor's type. CIFAR is also *not* amplitude-self-similar. Detail energy triples from the finest height to the coarsest, and kurtosis falls from 9.4 to 4.2. These images are downsampled, which suppresses fine detail. Finally, the class information in the thumbnail pyramid: a 2×2 thumbnail of the whole image gives 0.30 (linear) and 0.38 (MLP), a 4×4 one 0.38 and 0.53, and the full image 0.39 and 0.54.

Table 19: CIFAR-10 calibration (training set). Left: per child height c ($c=0$: 2×2 pixel patches), mean Haar detail energy, mean kurtosis of the nine coefficients, and residual correlation of log detail energies given the parent thumbnail, for siblings and for adjacent cousins. Right: test accuracy from the $k\times k$ block-average thumbnail.

c	energy	kurtosis	sibling	adj. cousin
0	0.050	9.4	0.25	0.45
1	0.086	7.7	0.23	0.39
2	0.118	5.9	0.22	0.40
3	0.150	4.2	0.35	–

k	linear	MLP
1	0.239	0.258
2	0.302	0.376
4	0.378	0.531
8	0.413	0.568
32	0.390	0.540

(I1) *Hidden context at every height.* IT(16) and IT(128) have lower test bits than IT(1) at *every* height (per-height bits per coordinate from truncated pyramids), not only in total.

(I2) *Heavy tails from the types* (Remark 49). Samples of IT($H>1$) reproduce the per-height kurtosis of the test set $(9.4, 7.7, 5.9, 4.2)$ better than IT(1), whose samples fall short at the fine heights.

(I3) *Readout.* Heads on the root of IT(1) stay at or below the 2×2 ceiling (0.30 linear, 0.38 MLP; Corollary 51). For $H>1$ no accuracy is predicted: the theory says what the root holds, not how well pixel-likelihood types align with classes. The *label need* of the Bayes head is predicted: it grows with H, the number of root types to name, and not with depth. With 1000 labels the Bayes head of IT(16) (160 label-kernel numbers) is within 0.02 of its accuracy with 50,000 labels, and IT(128) (1280) is further from it.

(I4) *Claim 1, likelihood* ($H=16$, N unlabelled images). Relative to $N=50{,}000$, the tied model's excess test bits per dimension falls like $1/N$, with magnitude $26032/(2N\cdot3072\ln2)=6.1/N$ within a factor of 3, for $N\geq256$. The untied model's excess is about $3.8\times$ larger at the same N (its parameter ratio).

(I5) *Claim 1, top of the tree.* At the top height (36 coordinates per image) the untied model's excess per coordinate is $\approx 24576/(2N\cdot36\ln2)=492/N$ bits, against the tied model's $\approx 6.1/N$, a ratio of about 80 (asymptotic, so tested for $N\geq4096$; below that the untied top overfits more still). The root representation follows: the logistic head (all labels) on the tied model trained on $N=1024$ images is within 0.02 of the same head at $N=50{,}000$, while the untied model at $N=1024$ is more than 0.02 below its own $N=50{,}000$ value.

(I6) *The price of tying on CIFAR.* Detail energy triples across heights, and the tied kernel can follow that only through the parent thumbnail's contrast. At $N=50{,}000$, where the untied model's variance term is negligible ($23.4/N\approx0.0005$ bits/dim), the untied model has the

lower test bits/dim, and its samples match the per-height detail energies of the data within 10%. The tied model's samples have a flatter profile (coarsest-to-finest energy ratio below the data's 3.0).

10.5 Results: one zero-label model, three uses

Run (`verify/imtree.py`, `verify/exp_cifar.py main`, Colab L4, 13 minutes for all four models). All 50,000 training images, no labels; Adam on the expected complete-data log-likelihood, 6000 steps of 128 images, cosine learning rate. *A correction:* the first evaluation subtracted the Jacobian constant of Lemma 46 once per batch of 500 test images instead of once per image. That shifted every test value by the same exact amount (0.3314 bits/dim in total, 0.3327 bits per coordinate per height), and it was caught because test came out below train. The numbers below are corrected; comparisons between models were never affected. The evaluator is now checked against image-by-image likelihoods (`verify/test_exp_cifar_eval.py`).

Table 20: CIFAR-10 test set, models trained with zero labels on 50,000 images. Density: bits/dim (8-bit, uniformly dequantized; an upper bound), and bits per detail coordinate at each child height c (lower is better). Readout: accuracy of heads on the *root state only* with P labels; Bayes head $= \sum_h \beta_R(h)\ell(y|h)$ (Prop. 50); logistic = multinomial logistic regression on $(\log \beta_R, t_R)$. Generation: kurtosis and energy of Haar details of 10,000 samples, per height (data: $9.4, 7.7, 5.8, 4.2$ and $0.050, 0.085, 0.118, 0.151$).

			density				Bayes head		logistic head	
model	params	bits/dim	c=0	1	2	3	P=10^3	$5{\cdot}10^4$	10^3	$5{\cdot}10^4$
IT(1)	3,127	5.923	−2.22	−1.79	−1.38	−0.76	0.100	0.100	0.276	0.302
IT(16)	26,032	4.984	−3.20	−2.61	−2.11	−1.62	0.163	0.167	0.297	0.333
IT(128)	253,952	**4.753**	−3.44	−2.83	−2.30	−1.81	0.161	0.183	0.255	**0.377**
IT(16) untied	99,760	4.967	−3.22	−2.62	−2.16	−1.78	0.245	**0.273**	0.318	0.361

samples	kurtosis, $c = 0..3$				energy, $c = 0..3$			
data	9.39	7.67	5.76	4.18	0.050	0.085	0.118	0.151
IT(1)	3.02	3.02	3.03	3.06	0.046	0.046	0.048	0.057
IT(16)	10.41	8.56	6.58	4.65	0.036	0.045	0.065	0.100
IT(128)	11.05	8.94	6.32	4.29	0.041	0.054	0.082	0.133
IT(16) untied	8.27	6.75	5.12	4.20	0.045	0.078	0.109	0.144

Reading Table 20. (I1) *Confirmed.* Hidden context types lower the test bits at every height, not only in total: IT(16) gains 0.98, 0.83, 0.73, 0.86 bits per coordinate over IT(1) at $c = 0..3$, and IT(128) gains a further 0.19–0.24 at each. In total, $5.92 \to 4.98 \to 4.75$ bits/dim. The state of an image region is not its thumbnail; the part that context adds is worth about one bit per coordinate at every scale. (I2) *Confirmed.* IT(1)'s samples are Gaussian at every height (kurtosis 3.0). With hidden types the samples reproduce the data's falling kurtosis profile ($10.4, 8.6, 6.6, 4.7$ and $11.1, 8.9, 6.3, 4.3$ against $9.4, 7.7, 5.8, 4.2$): the heavy tails of natural images come out of the hidden part of the state, as a scale mixture, and nothing else in the model produces them. (I3) *Confirmed where it predicts.* IT(1)'s logistic head reaches 0.302, exactly the 2×2 thumbnail's linear accuracy (0.302, Table 19); its Bayes head is at chance, since it has one type. The label need grows with H: from 10^3 to $5 \cdot 10^4$ labels the Bayes head of IT(16) moves by 0.005, that of IT(128) by 0.023. On the unpredicted part: the root is a weak representation of the class. The best head, 0.377 (IT(128),

logistic), equals the 2×2 thumbnail's MLP ceiling and sits below a linear classifier on raw pixels (0.390). The reason is visible in the root belief: its entropy is 0.13–0.39 nats, so the whole image selects essentially one root type. What the head sees is the thumbnail plus a hard clustering of images by the type that best explains their pixels. (I6) *Confirmed, one margin missed.* At $N = 50{,}000$ the untied model has the lower test bits/dim (4.967 against 4.984), and the difference lives at the top: 0.01–0.05 bits per coordinate at $c = 0..2$, 0.17 at $c = 3$, the object scale. Its sample energies are within 5–11% of the data at every height (the 10% margin is missed at $c = 0$). The tied model's profile is flatter, as predicted, but only slightly (ratio 2.8 against 3.0): the parent's contrast carries most of CIFAR's scale drift. *An unpredicted finding with a clear reading:* the untied top also makes the root far more class-informative (Bayes head 0.273 against 0.167 at the same H). The top of a CIFAR image is where tying breaks (largest bias at $c = 3$), and a free top kernel spends its types on object-scale structure. This is partial tying (§6) observed on real data: the root is exactly the level with no context of its own.

Samples (Figure 1). Scene-like colour layouts (sky above, foliage or ground below), flat regions and textured blobs, and strong blockiness along parent boundaries. The blockiness was predicted by the calibration: adjacent cousins, which meet across a parent boundary, are more dependent than siblings (0.39–0.45 against 0.22–0.35), and a tree can carry that dependence only through their common ancestor's type.

10.6 Results: depth adds data on CIFAR

Run (`verify/exp_cifar.py sweep`, Colab L4, 20 minutes). IT(16), tied and untied, trained with zero labels on the first N images of a fixed random order ($N \leq 1024$: two disjoint subsets), 3000 steps each, same schedule for every N. The reference for "excess" is the $N = 50{,}000$ fit of the same run, same schedule. (A first launch of this sweep was killed half-way by the session audit of the parallel main run, which treated the live sweep session as a ghost. The runner now stops only sessions whose owning job is gone. The rerun reproduced every completed fit to three decimals.)

Table 21: Claim 1 on CIFAR-10. Test bits/dim and bits per coordinate at the top height ($c = 3$, the four children of the root, 36 coordinates per image), for IT(16) trained on N unlabelled images; two subsets where shown. Excess is relative to $N = 50{,}000$. Predictions (Prop. 52): tied excess $6.1/N$ bits/dim and $\approx 6.1/N$ bits per coordinate at every height; untied excess 3.8× the tied total, and $492/N$ bits per top coordinate. Last column: logistic head on the root state, all 50,000 labels.

	bits/dim		excess bits/dim		excess, top (c=3)		root head
N	tied	untied	tied (pred.)	untied	tied	untied (pred.)	tied / untied
64	5.414, 5.416	29.9, 109	0.301 (0.095)	25, 104	0.39	36, 1647 (7.7)	0.309 / 0.308
256	5.182, 5.174	5.348, 5.353	0.064 (0.024)	0.258	0.08	7.6 (1.9)	0.319 / 0.314
1024	5.132, 5.122	5.133, 5.132	0.013 (0.006)	0.040	0.00	0.48 (0.48)	0.322 / 0.329
4096	5.128	5.098	0.014 (0.002)	0.005	0.02	0.09 (0.12)	0.333 / 0.345
16384	5.114	5.093	0.000 (0.000)	0.000	−0.01	0.01 (0.03)	0.335 / 0.345
50000	5.114	5.093	–	–	–	–	0.338 / 0.347

Reading Table 21. (I4) *Confirmed.* The tied excess falls like $1/N$: by 4.7 and 4.9× per fourfold increase from $N = 64$ to 1024. Its size is 2.7× and 2.2× the Wilks value at $N = 256$ and 1024. That is inside the predicted factor of 3, and it reads as $k^* \approx 2.5k$: the model is misspecified, so the effective number of parameters $\mathrm{tr}(J^{-1}I)$ exceeds the count. From $N = 1024$ on, the excess is below the optimization noise (≈ 0.01 bits/dim, the spread between the two subsets). The untied

excess is 4.0× and 3.1× the tied one at $N = 256$ and 1024, against 3.8 predicted. At $N = 64$ the untied model is non-asymptotic and collapses (30 and 109 bits/dim), while the tied model loses 0.3. (I5) *Confirmed for the density, refuted for the head.* At the top height, the untied model's excess matches the parameter-free $492/N$: 0.45–0.50 against 0.48 bits per coordinate at $N = 1024$, and 0.09 against 0.12 at 4096. The tied model's top excess is already below the noise floor at $N = 1024$, a ratio of more than 20. Below $N \approx 24576/36 \approx 700$ the untied top has fewer data coordinates than parameters and overfits beyond any asymptotic estimate (7.6 bits per coordinate at $N = 256$, against 1.9). But the predicted consequence for the root head did not appear. The untied head at $N = 1024$ is only 0.016–0.020 below its $N = 50{,}000$ value, not more than 0.02; even at $N = 64$, with a collapsed density, it is 0.31. The head reads an almost hard clustering of images (§10.5), and a badly estimated top kernel still yields a similar clustering. Since the root holds little class information to begin with, there is little for estimation error to destroy. *Bias against variance.* The two curves cross between $N = 1024$ and 4096. Below that the tied model wins, since the untied top pays for its parameters (Proposition 52). Above it the untied model wins by the tying bias of CIFAR, 0.021 bits/dim, concentrated at the top (0.19 bits per coordinate at $c = 3$). Equating the untied excess $\approx 2.5 \cdot 23.4/N$ with that bias predicts a crossover at $N \approx 2800$, inside the observed interval. *Optimization, not data, is now the limit.* The $N = 50{,}000$ fit with 6000 steps (Table 20) reaches 4.984 bits/dim, against 5.114 with 3000 steps: a gain of 0.13, ten times the estimation excess at $N = 1024$. From about a thousand unlabelled images on, the tied kernel is limited by optimization and by the model class, not by the amount of data.

10.7 What CIFAR-10 teaches

1. *The theory carries to real images, where it made predictions.* The state of a region is (thumbnail, hidden context); the thumbnail is parent-determined and needs no codebook; the hidden part is worth one bit per coordinate at every scale and carries the heavy tails (I1, I2). Depth adds data as the parametric accounting says: the tied kernel's excess is $\approx 2.5k/(2N)$ spread over all scales, while an untied top pays the predicted $492/N$ bits per coordinate, more than 20× the tied model's (I4, I5). About 10^3 unlabelled images give the kernel that $5 \cdot 10^4$ give.

2. *The root is not a class representation, and the theory says why.* A linear head on the root is Bayes-optimal only if the class is a child of the root state (Proposition 50). A likelihood-trained tied root is a 2×2 thumbnail plus one nearly certain context type, and reaches 0.30–0.38. The untied top did better (0.27 against 0.17 for the Bayes head at equal H): the top is where CIFAR stops being self-similar, and a root that is free to differ from the other levels spends its types on object-scale structure. The next theoretical object is therefore the *semantic top*: partial tying (§6) for continuous trees, with the height at which tying ends estimated from the per-height tying bias (here $c = 3$). Theorem 11 already says the untied top needs labels. The question is whether, given labels only there, the root becomes a class representation while everything below stays zero-label and tied.

3. *Seams are the tree's own limit.* Adjacent cousins are more dependent than siblings on CIFAR, and the samples are blocky along parent boundaries. A single quadtree cannot fix this. The recursive grid of the original notes allows "diagonal" coupling between grids that are not parent and child. That is the structural extension these data ask for, and it will cost exactness of sum–product, which the theory must price before it is built.

4. *Claim 2 has not been tested on CIFAR.* The head-to-head of §8.9 (same architecture, local EM against root-label backpropagation) is the obvious next run. Given item 2, it should be

run once the root carries class information.



Figure 1: Left: CIFAR-10 test images. Right: samples (top-down, exact ancestral sampling) from the four models of Table 20, all trained with zero labels.

10.8 The semantic top: the label as one more child of the root

In §10.5 the label was attached *after* learning: the model was fitted with zero labels and the label kernel ℓ was then fitted on a frozen root. The model of Proposition 50 is a joint one, $p(X, y)$, with y a child of the root. Its maximum-likelihood fit on a mix of labelled and unlabelled images uses the same generalized EM. The label leaf is observed where it is known and summed out where it is not. Nothing else is added, so the rule stays pure. Partial tying (§6) says where such a top should differ from the rest of the tree. This subsection works out what labels can and cannot do in that model, before any run.

Definition 53 (semantic top $\mathrm{ST}(H, c^*)$)**.** $\mathrm{IT}(H)$ in which the kernels of parents at heights $c \geq c^*$ are untied (one copy per height), the kernels below c^* are tied, and the label $y \in \{1, \dots, C\}$ is a fifth child of the root with kernel $\ell(y \mid h_R)$. $c^* = L + 1$ is the tied model, and $c^* = 1$ the untied one. It is trained by generalized EM on N images, of which P carry their label.

Proposition 54 (price of semantics)**.** *Let $\hat{\theta}_0$ maximize the pixel log-likelihood $\mathcal{L}_X(\theta) = \sum_n \log p_\theta(X_n)$, and let $\hat{\theta}$ maximize the joint log-likelihood $\mathcal{L}_X(\theta) + \sum_{n \leq P} \log p_\theta(y_n \mid X_n)$. Then*

$$0 \;\leq\; \mathcal{L}_X(\hat{\theta}_0) - \mathcal{L}_X(\hat{\theta}) \;\leq\; \sum_{n \leq P} \big(\log C + \log p_{\hat{\theta}}(y_n \mid X_n) \big) \;\leq\; P \log C.$$

On CIFAR-10 the pixel likelihood lost by learning with all labels is at most $\log_2 10/3072 = 0.0011$ *bits/dim.*

Proof. Let θ_0' be $\hat{\theta}_0$ with the uniform label kernel. Its pixel marginal is that of $\hat{\theta}_0$, and $\log p_{\theta_0'}(y \mid X) = -\log C$. The joint optimality of $\hat{\theta}$ against θ_0' gives the middle inequality; the others follow from the optimality of $\hat{\theta}_0$ and from $\log p(y \mid X) \leq 0$. □

This works in both directions. Semantics cost nothing in density. But labels can move the model only within the set of models whose pixel likelihood is within $H(Y) \leq \log C$ nats per labelled image of the optimum. A class-informative root can be selected by labels only if one already exists among the near-optimal pixel models. On an image of 3072 coordinates the label is worth one thousandth of a bit per coordinate, and it can win only where the pixels are indifferent. The root type is such a place: the root's own terms are a few bits per image, and they are the only terms that the root type alone decides.

The next statement says how far below the root that indifference extends. For positive vectors u, u' on the types let $d_H(u,u') = \log\max_h \frac{u(h)}{u'(h)} + \log\max_h \frac{u'(h)}{u(h)}$ (Hilbert's projective metric; it is unchanged by normalization and by multiplying both vectors by the same positive vector). For a positive kernel T let $\Delta(T) = \max\log\frac{T[a,b]\,T[a',b']}{T[a,b']\,T[a',b]}$ and $\kappa(T) = \tanh(\Delta(T)/4) < 1$.

Theorem 55 (label reach)**.** *Fix an image X and a label y. Let q_v and q_v^y be the posteriors of the type at node v without and with the label observed. Let v be at depth d (distance from the root), and let $T^{(1)},\dots,T^{(d)}$ be the kernels on the edges of its path. Put $\lambda_y = d_H(\ell(y\mid\cdot),\mathbf{1}) = \operatorname{osc}_h \log\ell(y\mid h)$ and $\epsilon_v = 1-\max_h q_v(h)$. Then*

$$d_H(q_v^y, q_v) \;\le\; D_v := \lambda_y \prod_{j=1}^{d} \kappa(T^{(j)}), \qquad \mathrm{TV}(q_v^y, q_v) \;\le\; \min\Big\{\tanh\frac{D_v}{4},\; \frac{\epsilon_v\, e^{D_v}}{1-\epsilon_v}\Big\}.$$

The joint posterior of v and its parent obeys the bound of the parent ($D_p = D_v/\kappa(T^{(d)})$).

Proof. The posterior of v is $q_v \propto (R_p\,T)\odot a_v$. Here a_v collects v's own emission and the messages from its children, and R_p is the belief of its parent p without v's message. Only R_p depends on the label. At the root, $R^y = R\odot\ell(y\mid\cdot)$, so $d_H(R^y,R) = \lambda_y$. A row vector times a positive kernel contracts d_H by $\kappa(T)$ (Birkhoff–Hopf). Multiplication by a_v, and by the messages of v's other children when passing to the next level, leaves d_H unchanged. Induction down the path gives the first bound. For probability vectors $\mathrm{TV} \le \tanh(d_H/4)$. For the second form, write $q_v^y \propto q_v\, r$ with $r(h_0) = 1$ at the mode h_0 and $r \in [e^{-D_v}, e^{D_v}]$. Then $1 - q_v^y(h_0) \le \epsilon_v e^{D_v}/(1-\epsilon_v)$, which bounds the total variation. The joint posterior is $\propto R_p(h)T[h,h']a_v(h')$, in which only R_p depends on the label, so the same argument applies with D_p. □

Corollary 56 (both rules see the label through the same reach)**.** *(a)* EM. *On a labelled image, the expected statistics of node v (its type and its parent pair) differ from those of the same image unlabelled by at most the bound of Theorem 55. The pixel statistics of every node of every image are label-free. (b)* Backpropagation from the root label. *For any parameter θ of the emissions and transitions,*

$$\partial_\theta \log p(y\mid X) = \sum_v \sum_h \big(q_v^y(h) - q_v(h)\big)\,\partial_\theta \log e_v(h) + \sum_v \sum_{h,h'} \big(\xi_v^y - \xi_v\big)(h,h')\,\partial_\theta \log T_v[h,h'],$$

with ξ_v the parent–child pair posterior and the root included as $e_R(h) = \pi(h)\,N(t_R; m_h, S_h)$ (Fisher identity for $p(X,y)$ and $p(X)$). Node v contributes at most $\mathrm{TV}(q_v^y, q_v)\operatorname{osc}_h \partial_\theta \log e_v(h)$ plus the same for its pair. So the label signal that reaches a node is the label reach in either rule.

This is the per-image counterpart of Theorem 41. There the root–leaf correlation decays at rate q_1 over the process; here the influence of an observed label decays at rate κ per edge inside the fitted model, and it also vanishes wherever the image already decides the type ($\epsilon_v \to 0$). The two rules differ in what else they use. EM also uses the label-free statistics of every node, and depth multiplies these (claim 1). Backpropagation from labels has only the reach. Three consequences follow for the design, and each is testable.

1. *Labels shape the top, and the top is where untying pays.* Below the depth where D_v or $\epsilon_v e^{D_v}$ is small, labelled and unlabelled images give the same statistics. There, the kernel is learned from unlabelled data at every height, and tying costs nothing. So c^* should be where the reach ends, and the untied heights are paid for by Proposition 52.

2. *Ceiling.* If the kernels below c^* are unchanged by the labels, the root belief, and hence the Bayes head, is a function of the *state grid* G_{c^*-1}. This consists of the thumbnails of the nodes at height c^*-1, which determine every coarser thumbnail and detail, and the messages those nodes receive from their children. No head of $\mathrm{ST}(H, c^*)$ beats the Bayes accuracy of y from G_{c^*-1}. Lowering c^* raises this ceiling and adds 4^{L-c^*+1}-fold more untied edges.

3. *Naming.* The root has no context, so its types are named by labels, and only by labels (Theorem 11). The label need is set by the number of top types to name and by the few parameters of the top that the labels move, not by depth.

10.9 Calibration and predictions for the semantic top (fixed before the run)

Calibration (`verify/exp_semtop.py cal`, one zero-label tied IT(16), 3000 steps, test 5.120 bits/dim; Table 22). The fitted kernel is nearly deterministic in the Birkhoff sense: $\kappa = 0.979$–0.987 at the four positions. The label kernel fitted on its root has $\lambda_y = 8.5$–12.1 nats, and the mean posterior uncertainty is $\epsilon = 0.11$–0.14 at every height (0.07 at the root). So $D_v \approx 9$ at every depth of a CIFAR tree, and *Theorem 55 is vacuous on CIFAR*: it allows a label to change any type in the image. It predicts no decay with depth over four levels, and we do not claim one. The label's actual reach is measured below, but not predicted. The class information in the state grids is the other half. An MLP on G_3 (the four quadrants: 4×4 thumbnail and their incoming messages) reaches 0.554, and on G_2 (16 regions) 0.622. The zero-label root reaches only 0.380 from the same model. The messages add class information beyond the thumbnails at every height, more so lower down (+0.05, +0.10, +0.12 for linear heads at $c = 3, 2, 1$). *The hidden context types of a zero-label tree are partly semantic*, and the loss happens at the last merge into the root.

Table 22: Semantic-top calibration: test accuracy of heads (all 50,000 labels, 5000 held out for model selection) on the state grid G_c of a zero-label tied IT(16): thumbnails only, messages only, and both. The root row uses $(t_R, \log \beta_R)$.

	nodes	thumbnails		messages		state G_c	
		linear	MLP	linear	MLP	linear	MLP
root	1	0.301	0.361	–	–	0.331	0.380
$c = 3$	4	0.379	0.538	0.286	0.305	0.432	0.554
$c = 2$	16	0.409	0.589	0.379	0.430	0.504	0.622
$c = 1$	64	0.411	0.582	0.451	0.502	0.529	0.640

Run. $\mathrm{ST}(16, c^*)$ for $c^* = 5$ (tied), 4 (root kernel untied) and 3 (top two untied), each with $P = 0$, 1000 and 50,000 labelled images among the $N = 50{,}000$, all trained from the same initialization by generalized EM on the joint likelihood (3000 steps). Heads: the joint Bayes head $\sum_h \beta_R(h)\ell(y \mid h)$ with the ℓ learned in training; for $P = 0$, the post-hoc Bayes head of §10.5.

(S1) *Price of semantics* (Proposition 54). For every c^*, the test bits/dim at $P = 50{,}000$ and at $P = 1000$ are within 0.01 of those at $P = 0$. The bound is 0.0011 at the optima; 0.01 is the optimization noise measured in §10.6.

(S2) *Label reach.* The bound of Theorem 55 holds at every node (it is a theorem; this checks the code). No quantitative prediction (see above).

(S3) *Ceiling* (Corollary 56, item 2). The joint Bayes head stays below the state-grid estimate plus 0.02 for the MLP's estimation slack: ≤ 0.57 for $c^* = 5, 4$ and ≤ 0.64 for $c^* = 3$.

(S4) *Labels select a semantic root.* The labels move only what the pixels are indifferent to, and the root type is such a place (Proposition 54). So at $P = 50{,}000$ the joint Bayes head exceeds the zero-label post-hoc Bayes head of the same c^* (0.167 tied, §10.5). The risky part: it exceeds the zero-label root's best linear head (0.33). The labels then produce a root that is more class-informative than any linear readout of the unlabelled one.

(S5) *Few labels name, many labels shape.* The pixel side of the root is paid by all N images, and the label side is gained on P of them. At $P/N = 0.02$ the labels cannot reshape the root: the joint Bayes head at $P = 1000$ is within 0.03 of the zero-label model's post-hoc Bayes head with 1000 labels. Semi-supervised EM with few labels names the unlabelled types; it does not make new ones.

(S6) *Untying the top helps the labels.* At $P = 50{,}000$ the joint Bayes head is non-decreasing from $c^* = 5$ to 4 to 3: an untied top can serve the labels without changing the kernel shared by all heights below.

10.10 Results: the semantic top

Run (`verify/exp_semtop.py run`, Colab L4, 10 minutes for the nine fits).

Table 23: ST$(16, c^*)$ on CIFAR-10, $N = 50{,}000$ images of which P are labelled, joint generalized EM. Test bits/dim; root heads (joint Bayes head with the ℓ learned in training; post-hoc Bayes head and logistic head on $(\log \beta_R, t_R)$ with 50,000 labels); root belief entropy (nats). Right: measured label reach, the mean $\mathrm{TV}(q_v^y, q_v)$ over test images (true labels) at each height; Theorem 55's bound is 0.12–0.39 at every height and was never violated.

			root heads				label reach (mean TV)				
c^*	P	bits/dim	joint	post-hoc	logistic	entropy	root	c=3	2	1	0
5	0	5.120	–	0.176	0.338	0.15					
	1000	5.121	0.169	0.182	0.337	0.15	0.024	$6 \cdot 10^{-4}$	$4 \cdot 10^{-5}$	$2 \cdot 10^{-6}$	$1 \cdot 10^{-7}$
	50000	5.120	0.167	0.175	0.334	0.11	0.028	$2 \cdot 10^{-3}$	$1 \cdot 10^{-4}$	$5 \cdot 10^{-6}$	$2 \cdot 10^{-7}$
4	0	5.116	–	0.221	0.336	0.29					
	1000	5.113	0.189	0.211	0.344	0.25	0.031	$2 \cdot 10^{-3}$	$1 \cdot 10^{-4}$	$7 \cdot 10^{-6}$	$5 \cdot 10^{-7}$
	50000	5.109	**0.247**	0.242	0.339	0.32	0.068	$4 \cdot 10^{-3}$	$2 \cdot 10^{-4}$	$1 \cdot 10^{-5}$	$8 \cdot 10^{-7}$
3	0	5.130	–	0.226	0.348	0.30					
	1000	5.117	0.210	0.229	0.356	0.31	0.032	$1 \cdot 10^{-3}$	$7 \cdot 10^{-5}$	$3 \cdot 10^{-6}$	$2 \cdot 10^{-7}$
	50000	5.115	**0.252**	0.252	0.355	0.37	0.068	$4 \cdot 10^{-3}$	$2 \cdot 10^{-4}$	$9 \cdot 10^{-6}$	$5 \cdot 10^{-7}$

Reading Table 23. (S1) *Confirmed in substance, with one margin missed in the harmless direction.* Labels never cost density: the largest increase is 0.002 bits/dim ($c^* = 5$, $P = 1000$), inside the theorem's 0.0011 plus noise. At $c^* = 3$ the labelled fits are 0.013–0.015 *better* than the zero-label fit, which is outside the stated 0.01. That zero-label fit is simply a worse optimum ($c = 2$ and $c = 3$ bits are both worse), so the optimization noise of the more untied model is larger than the 0.01 measured for the tied one. (S2) *Confirmed* (no violation at any node). See below for what the measured reach shows. (S3) *Confirmed, but not binding*: every head is far below the ceilings (0.57, 0.64). (S4) *Refuted for the tied model, confirmed for the untied tops, and the risky part refuted.* With all labels, the tied model's joint head is 0.167, no better than its zero-label post-hoc head. The untied tops gain: 0.221 → 0.247 at $c^* = 4$ and 0.226 → 0.252 at $c^* = 3$. No joint head

approaches the 0.33 of a linear readout of the unlabelled root. (S5) *Confirmed.* At $P = 1000$ the joint head is within 0.003–0.016 of the zero-label post-hoc head, and always below it. Few labels name; they do not shape. (S6) *Confirmed*: $0.167 \leq 0.247 \leq 0.252$.

Why (S4) failed: the premise was wrong, and the calibration already showed it. Proposition 54 says labels can move only what the pixels are indifferent to. We assumed the root type was such a place. It is not: the root belief has entropy 0.11–0.37 nats, so the pixels (the root thumbnail and the four quadrant types) already decide the root type with a margin of several nats per image. The label pays at most λ_y, and the learned label kernels have $\lambda_y \approx 1$–3 nats: the label kernel is fitted to root types that are not class-pure, and so it cannot pay for making them purer. This is a self-consistent fixed point of joint EM, and the calibration's root entropy (0.15) was the number that should have predicted it. Per image the statement is: an image's root type can follow its label only if its pixel margin between root types is below the label kernel's spread. Pure likelihood with the label as a root child therefore cannot create semantics at a root that the pixels have already decided. Untying the top loosens the root (entropy $0.15 \to 0.3$), and only then do the labels move it a little.

What the measured reach shows. The label's influence falls by a factor of 15–25 per level: from TV ≈ 0.03–0.07 at the root to $\approx 10^{-7}$ at the 2×2 patches. In d_H the per-edge ratio is 0.03–0.06, nearly constant with depth (for $c^* = 5$, $P = 50{,}000$: 2.9, 0.10, 0.0055, 0.00028, $1.4 \cdot 10^{-5}$). The kernel's own Birkhoff constant is $\kappa \approx 0.98$, so the contraction does not come from the kernel. It comes from the certainty of the parent. If the parent's belief R puts mass $1 - \epsilon$ on a type h_0 and the label multiplies it by weights within $e^{\pm d}$, then with $\rho = \max_{h,h'} T[h, h']/T[h_0, h']$,

$$d_H(R^y T,\, RT) \;\leq\; \log \frac{1 + e^d a}{1 + e^{-d} a}, \qquad a = \frac{\epsilon\rho}{1 - \epsilon},$$

which is $\approx 2\sinh(d)\, a$ for small a: the downward perturbation is damped by the parent's uncertainty $\epsilon \approx 0.1$ (proof: only the non-modal part of R is reweighted relative to the mode, and it enters RT with weight at most a relative to the modal row). We did not measure ρ, so the constant is not checked; the mechanism matches the observed order of magnitude. For claim 2 this is Corollary 56(b) measured: *per image, the root label's signal reaching a node falls about twentyfold per level on CIFAR*, so even summed over the 4^d nodes at depth d it falls about fivefold per level. Backpropagation from root labels can teach, on its own, only the top one or two levels of this tree. Below that, the kernel must come from unlabelled data, which is what tying and claim 1 provide.

What the semantic top teaches. The class information is present in the zero-label tree (0.55 and 0.62 from the state grids at $c = 3$ and 2, with the hidden types contributing beyond the thumbnails) and is lost at the last merge, where a near-certain discrete root type is chosen by pixel likelihood. Labels observed at that root, with the pure joint likelihood, cannot get it back: they are too weak against a decided root, and too weak to reach below it. The theory now says what a semantic top needs. Its state must stay uncertain enough under the pixels, meaning a margin below λ_y, for labels to shape it. It must also carry the grid's information into the root rather than a single type chosen by pixels. Both are properties of the top's state space, not of the learning rule.

10.11 A context for the root: views of one scene

Diagnosis. Every hidden type below the root is shaped by its context: it must carry what its siblings and parent need (Theorem 21, Proposition 47). The root has no context. Its type is paid only by its own children: it is whatever makes the root thumbnail and the four quadrant states conditionally independent. Pixel likelihood spends it on the largest such dependence, which on CIFAR is the global look of the image. The calibration shows this: the root type adds 0.02–0.03 to

the class accuracy of the whole image's 2×2 colour thumbnail (Table 22). There are two further defects: the root sees the image only through a 16-way merge of four near-certain quadrant types, and a tied root means what a pixel patch means. A label cannot repair this. It is a context worth at most $\log C$ per image, and the model can neutralize it at no pixel cost (Proposition 54). The root needs a context made of pixels.

Definition 57 (scene grammar $\mathrm{SG}(H, M)$ with K views)**.** A scene type $z \in \{1, \dots, M\} \sim \pi$ is the common parent of K views, each a whole image tree. Every node of a view has state (t, h, z), with z copied from parent to child (parent-determined and hidden). Given z, a view is $\mathrm{IT}(H)$ with root prior $\pi_z(h)$ and type kernel T_z (tied across heights); the emissions are shared by all z. So z selects a grammar over shared types, and the types are its words. The label, when observed, is a child of the scene node with kernel $\ell(y \mid z)$. Given z the views are independent trees, so sum–product is exact at M times the cost of the type passes. With one view the scene node is a mixture over M tied grammars.

The scene type sees a view through its whole tree: $\log P(z \mid X) = \log \pi(z) + \log p_z(X) + \text{const}$, and $\log p_z(X)$ sums the transition evidence of all 340 edges. There is no 16-way merge at the top, which removes the second defect. The first defect is what the views are for.

Proposition 58 (agreement identity)**.** *For K views and $w_k(z) = P(z \mid X^k)$, the single-view posteriors under the same model,*

$$\log p(X^1, \dots, X^K) = \sum_k \log p(X^k) + \underbrace{\log \sum_z \pi(z) \prod_k \frac{w_k(z)}{\pi(z)}}_{\textit{agreement}} .$$

With the label as a view ($p(y \mid z) = \ell(y \mid z)$), the agreement term is $\log \sum_z w(z)\ell(y \mid z) = \log p(y \mid X)$.

Proof. $p(X^1, \dots, X^K) = \sum_z \pi(z) \prod_k p_z(X^k)$, and $p_z(X^k) = w_k(z)p(X^k)/\pi(z)$ by Bayes. □

Corollary 59 (leverage of a context)**.** *(a)* Label. *Setting ℓ uniform makes the agreement term $-\log C$, whatever z does, at no pixel cost. So a label can move z by at most $\log C$ per labelled image, and the model can refuse it cheaply. This is what happened in §10.10, where the learned ℓ had $\lambda_y \approx 1$–3 nats. (b)* Views. *The agreement term is at most $(K-1)\log(1/\min_z \pi(z))$, and it is attained when the views' posteriors agree and are sharp. If the views' posteriors are sharp but disagree, it is about minus the smaller evidence margin, without bound. The model has only two ways out. Either z is a function of what the views share, or z is uninformative ($w_k = \pi$), which gives up the whole single-view gain of the mixture. So views can move z by up to the entire value of z to the pixels.*

Proof. (a) $\log \sum_z w(z)/C = -\log C$. (b) $\sum_z \pi \prod_k (w_k/\pi) \le \max_z \pi(z)^{-(K-1)} \sum_z w_1(z) \le \min_z \pi(z)^{-(K-1)}$. If w_1 and w_2 put mass $\ge 1 - \epsilon$ on different types, the sum is at most $2\epsilon/\min \pi$, and $\log \epsilon$ is minus the evidence margin. □

What z becomes, if it agrees across views, is fixed by the view family.

Theorem 60 (the scene type is the view invariant)**.** *Let S be a random scene, $g_1, g_2, \dots$ i.i.d. from ν and independent of S, and $X^k = g_k \cdot S$. Let $O(S)$ be the law of $g \cdot S$ given S. Then: (a) the views are conditionally i.i.d. given $O(S)$; (b) if some latent Z makes them conditionally i.i.d. with law Q_Z, then $Q_Z = O(S)$ almost surely: the views determine $O(S)$, and a scene type*

distinguishes nothing finer than $O(S)$; (c) if ν is Haar measure on a compact group G acting on scenes, then $O(S) = O(S')$ iff $S' \in G \cdot S$: the scene type is the G-orbit, the content of the scene that the transformations do not change.

Proof. (a) Given S, the g_k are i.i.d. (b) The empirical measure of $X^1, \dots, X^n$ converges almost surely (on a countable convergence-determining class) to Q_Z, because the views are conditionally i.i.d. given Z, and to $O(S)$ by (a). (c) If $S' = h \cdot S$ then $g \cdot S' = (gh) \cdot S$, which has the law of $g \cdot S$ by invariance of Haar measure. Otherwise the two laws live on the disjoint closed orbits $G \cdot S$ and $G \cdot S'$. □

With finitely many scene types, maximum likelihood chooses among partitions of the invariant content: the agreement term (Corollary 59) requires z to be (nearly) a function of $O(S)$, and the single-view term picks, among such functions, the one worth the most pixels. Three consequences follow, each testable.

1. *The family decides the semantics.* Crops and flips leave the global colour shared, so z may carry it. With independent colour jitter, colour is not shared, and z must drop it. Class enters z only as part of the invariant, and only if it is worth pixels. The theory does not promise that it is; it says this is the only route.

2. *Depth adds data at the top.* The scene node has K children, so its kernel is learned from K views per scene, as every other node's is learned from its siblings (Theorem 26).

3. *Labels only name.* With z fixed by the views, labels fit ℓ, which has $M(C-1)$ numbers (Theorem 11).

Predictions (fixed before the run). Models, all with zero labels on the 50,000 training images: R, the tied IT(16) (the old root); A, SG(16, 16) with one view and no transformation (no merge bottleneck, no context); B, two views under random crops (4-pixel reflect padding) and flips; C, two views under crops, flips and independent colour jitter (brightness, contrast, saturation, channel gains, 20% grey); D, as C with four views. 3000 steps of 128 scenes each. The state of a test image is its single-view posterior $w(z)$ (R: the root belief).

(W1) *Agreement produces invariance* (Corollary 59, Theorem 60). The mean $\mathrm{TV}(w(gX), w(X))$ under random crops and flips is at most half of A's for B, C and D. Under colour jitter it is at most half of B's for C and D.

(W2) *The family decides the semantics.* The R^2 of an image's mean colour from its state is at most half as large for C as for B.

(W3) *Without context the root stays a pixel clustering.* A's state carries the global look (colour R^2 at least that of R), and its post-hoc Bayes head is at most 0.28: removing the merge alone does not make the root semantic.

(W4) *Density.* A contains R ($M = 1$), so A's test bits/dim is at most R's +0.01.

(W5) *Class, the risky one.* With all labels naming the scene types, the Bayes head of C is at least that of B, at least 0.30 (above every root so far, ≤ 0.25), and D's is at least C's.

(W6) *Labels only name.* For B, C, D the Bayes head with 1000 labels is within 0.02 of that with 50,000.

10.12 Results: views of one scene

Run (`verify/scenetree.py`, `verify/exp_views.py`; checks in `verify/test_scenetree.py`: brute force, Fisher identity for $K = 1, 2, 3$ with and without labels, the agreement identity, and $M = 1$ equal to IT; Colab L4, 12 minutes). One change was made after the predictions were fixed and before the run: a smoke test showed one scene type taking 99% of the images at initialization, and a mixture component that starts with no responsibility never receives a gradient. The M kernels therefore start from a shared kernel plus a small (0.005) per-type perturbation, which gives every type about $1/M$ of the images at the start.

Table 24: Views of one scene, CIFAR-10, zero labels in training. Types used: scene types with $\geq 5\%$ of the test images. Heads on the state: post-hoc Bayes with 10^3 and $5 \cdot 10^4$ labels, logistic on the state, and logistic on the state and the root thumbnail. Invariance: mean $\mathrm{TV}(w(gX), w(X))$ for crops and flips, for colour jitter, and between two different images (scale). Colour: R^2 of the image's mean colour from the state (D: least squares ill-conditioned, not reported).

model	types used	bits/dim	Bayes 10^3	Bayes $5{\cdot}10^4$	logistic state	logistic $+t_R$	entropy	invariance TV crop	invariance TV colour	invariance TV other	colour R^2
R: IT(16) root	2	5.120	0.166	0.176	0.294	0.337	0.15	0.087	0.231	0.421	0.945
A: $K = 1$	3	5.108	0.190	0.190	0.230	0.349	0.03	0.141	0.216	0.647	0.126
B: $K = 2$, crop	3	5.114	0.190	0.190	0.235	0.347	0.04	0.154	0.237	0.665	0.127
C: $K = 2$, crop+colour	2	5.128	0.148	0.154	0.185	0.316	0.01	0.054	0.119	0.341	0.128
D: $K = 4$, crop+colour	2	5.119	0.152	0.158	0.200	0.324	0.01	0.065	0.105	0.394	–

Reading Table 24. (W1) *Refuted in substance.* B is no more crop-invariant than A (0.154 against 0.141). C and D pass the stated test (crop TV 0.054, 0.065 against A's 0.141; D's colour TV 0.105 against B's 0.237; C's 0.119 misses by 0.001). But they do it by using fewer types: relative to the TV between two different images, the invariance of C and D is no better than B's (colour: 0.35, 0.27 against 0.36). (W2) *Not a test.* No scene state carries colour ($R^2 \approx 0.13$ for A, B and C). In this design colour lives in the root type of each view, not in z, so the prediction could not discriminate. (W3) *Half refuted.* A's state does not carry the global look ($R^2 = 0.13$, against 0.945 for the old root). Its Bayes head is $0.190 \leq 0.28$, as predicted. (W4) *Confirmed*: A 5.108 against R 5.120. (W5) *Refuted.* C (0.154) is below B (0.190), far below 0.30, and D equals C. (W6) *Confirmed, trivially*: with two or three types in use there is little to name.

What happened, and what it says about the root. (i) *The old root is a colour code.* Its state predicts the image's mean colour with $R^2 = 0.945$ and uses essentially two of its sixteen types (73% and 21% of the images). This confirms the diagnosis at the start of §10.11 directly. (ii) *A global discrete variable is decided by hundreds of nats.* Once z sees every edge, its single-view posterior has entropy 0.01–0.04 nats. Generalized EM on such hard posteriors keeps the few components that win early: two or three of sixteen in every run. The near-symmetric start delays this but does not prevent it. (iii) *The views pruned; they did not create.* Corollary 59(b) gives the model two ways out of disagreement: an invariant z, or an uninformative one. With colour jitter the model took the second, keeping two scene types. We had argued that this exit is expensive because it gives up the single-view gain of the mixture. That gain turned out to be small: A is only 0.012 bits/dim better than R. Views can remove distinctions that are not shared, but they cannot make the likelihood want a distinction, such as the class, that is worth few pixels. Theorem 60 says what z may be; the single-view term decides what it is, and on CIFAR it prefers very little. (iv) *Where the class is.* Linear heads on the zero-label state grids reach 0.43, 0.50, 0.53 at $c = 3, 2, 1$ (Table 22), against

0.39 for a linear head on the raw pixels: the unlabelled tree's region states are better linear features than pixels. Every attempt to push this into *one* variable at the top, whether a root type, a labelled root, or a scene type shared by views, has produced a coarse clustering by bulk statistics. The class is carried by the *configuration* of region states, and a strong root must keep that configuration rather than choose among a few pixel prototypes.

10.13 A root that keeps the configuration: the state grid

Every failure above has the same cause: some variable at the top was chosen by pixel likelihood, and the pixels chose bulk statistics. The alternative is to choose *nothing* by pixel likelihood above some height c. Below c, every node has context (siblings and a parent), and the zero-label tree is what the theory says it should be. Above c, the class reads the whole grid of region states.

Definition 61 (grid-parent model GP(c))**.** Keep the tied tree below height c. Replace everything above by the class: $y \sim p(y)$, and given y the $n_c = 4^{L-c}$ nodes v at height c are independent, with type $h_v \sim \rho_v(h \mid y)$ and thumbnail $t_v \sim N(\mu_{v,y}, \Sigma_v)$. The head is *untied* if ρ_v, μ_v, Σ_v depend on the position v, and *tied* if they do not.

Proposition 62 (grid readout)**.** *Let $e_v(h) = p(X_{<v} \mid h_v = h, t_v)$ be the message that grid node v receives from its subtree (computed by the zero-label tree), and $\tilde{e}_v = e_v / \sum_h e_v(h)$. Under* GP($c$)*,*

$$\log P(y \mid X) = \log p(y) + \sum_v \Big[\log \sum_h \rho_v(h \mid y)\, \tilde{e}_v(h) + \log N(t_v; \mu_{v,y}, \Sigma_v) \Big] + \text{const}(X).$$

If $\tilde{e}_v$ puts mass $1 - \epsilon_v$ on one type h^*, then $0 \le \log \sum_h \rho_v(h \mid y) \tilde{e}_v(h) - \sum_h \tilde{e}_v(h) \log \rho_v(h \mid y) \le \log\big(1 + \epsilon_v \frac{\max \rho}{\rho(h^*)}\big) + \epsilon_v \log \frac{\rho(h^*)}{\min \rho}$. So, up to that error, the Bayes log-odds are linear *in the features $(\tilde{e}_v, t_v)_v$: a linear head on the grid is the Bayes rule of* GP(c)*.*

Proof. Given y the grid nodes are independent, and below each of them the tree is the zero-label tree, so $p(X \mid y) = \prod_v \sum_h \rho_v(h \mid y) N(t_v; \mu_{v,y}, \Sigma_v)\, e_v(h)$. Normalizing e_v changes only a factor that does not depend on y. The quadratic terms in t_v cancel between classes because Σ_v is shared. The lower bound is Jensen's inequality. For the upper bound, the sum is at most $\rho(h^*) + \epsilon_v \max \rho$, and $\sum_h \tilde{e}_v \log \rho \ge (1 - \epsilon_v) \log \rho(h^*) + \epsilon_v \log \min \rho$. □

Nothing above height c is chosen by pixels, so the failure of §10.10 and §10.12 cannot occur. The labels choose which regions, which types and which colours matter. The price is that labels now fit more numbers than a root kernel.

Corollary 63 (who pays for what)**.** *(a)* Labels. *The untied head has $d_c = n_c(H + 12)(C - 1)$ free numbers, and its expected excess log-loss from P labels is $\approx d_c/(2P)$ (Proposition 52). So its label need grows fourfold per level of refinement. The tied head has $d = (H + 12)(C - 1)$ at every c, since it reads the grid as a bag of region states and gives up position. This is self-similarity applied to the readout. (b)* Unlabelled images. *The features $(\tilde{e}_v, t_v)$ are made by the zero-label tied kernel, which $\approx 10^3$ unlabelled images already fix (§10.6). So the readout's accuracy is fixed by $\approx 10^3$ unlabelled images as well (claim 1, now for classification). (c)* Same architecture, two rules *(claim 2). Local EM learns the kernel from unlabelled images, and labels fit only the head. Backpropagation of the label loss through the same tree must also learn the kernel (*26,032 *numbers) from the labels, and the label's signal reaches the nodes below the grid only through the label reach, which fell about twentyfold per level on CIFAR (§10.10).*

Predictions (fixed before the run). Features from the zero-label tied IT(16) (3000 steps, N unlabelled images), at $c = 3, 2, 1$ (4, 16, 64 regions). Heads: untied and tied logistic (conditional maximum likelihood of GP(c)), and the generative GP(c) fit by maximum likelihood (the Bayes grid head: $\rho_v(h \mid y)$ from soft counts of $\tilde{e}_v$, class means and a pooled covariance for t_v). Labels $P \in \{100, 300, 10^3, 3 \cdot 10^3, 10^4, 5 \cdot 10^4\}$. $P_{0.02}$ denotes the smallest of these at which a head is within 0.02 of its own accuracy at $P = 5 \cdot 10^4$.

(G1) *Label need grows with the grid* (Corollary 63(a)). For the untied logistic head, $P_{0.02}$ is non-decreasing from $c = 3$ to 2 to 1; at $c = 3$ it is at most 10^4, and at $c = 1$ it is $5 \cdot 10^4$ ($d_1 = 17{,}920$).

(G2) *The tied head is label-cheap.* At every c its $P_{0.02}$ is at most 3000.

(G3) *Depth adds data to the readout* (Corollary 63(b)). The untied head at $c = 2$ with all labels, on features from $N = 1024$ unlabelled images, is within 0.02 of the same head on features from $N = 50{,}000$.

(G4) *Claim 2* (Corollary 63(c)). At $c = 2$, EM features with the untied logistic head against backpropagation of the cross-entropy through the same tree and head from the same initialization (learning rate best of three on 5000 held-out labelled images, early stopping on them, neither counted in P). EM is at least as accurate at every $P \leq 10^4$, and ahead by at least 0.05 at $P = 1000$.

(G5) *The grid beats the root, even generatively.* The Bayes grid head at $c = 2$ with all labels exceeds 0.30, above every Bayes head on a single top variable so far (≤ 0.25).

10.14 Results: the state grid

Run (`verify/exp_grid.py`, Colab L4, 12 minutes).

Table 25: Readouts of the zero-label tied IT(16)'s state grid, CIFAR-10 test accuracy against the number of labels P. Features from $N = 50{,}000$ unlabelled images. Last row: backpropagation of the cross-entropy through the same tree and untied head ($c = 2$) from the same initialization, learning rate best of three and early stopping on 5000 further labelled images; its largest P is 45,000.

	head	$P = 100$	300	10^3	$3 \cdot 10^3$	10^4	$5 \cdot 10^4$
$c = 3$ (4 regions)	untied	0.221	0.232	0.318	0.387	0.420	0.434
	tied	0.199	0.245	0.313	0.336	0.339	0.341
	Bayes grid	0.247	0.319	0.362	0.377	0.384	0.386
$c = 2$ (16 regions)	untied	0.243	0.290	0.336	0.379	0.462	**0.505**
	tied	0.214	0.286	0.333	0.351	0.360	0.365
	Bayes grid	**0.259**	**0.337**	**0.382**	0.396	0.403	0.409
$c = 1$ (64 regions)	untied	0.249	0.342	0.366	0.388	0.419	**0.525**
	tied	0.225	0.306	0.364	0.380	0.394	0.388
	Bayes grid	0.239	0.337	0.370	0.389	0.397	0.407
$c = 2$	backprop	–	0.283	0.373	**0.425**	**0.475**	0.483

Reading Table 25. (G1) *Confirmed.* The untied head's $P_{0.02}$ is 10^4 at $c = 3$ and $5 \cdot 10^4$ at $c = 2$ and $c = 1$: the finer the grid, the more labels, as d_c says. (G2) *Confirmed.* The tied head is within 0.02 of its limit at $P = 3000$ at every c. It pays for this with a lower limit (0.34–0.39): position matters. (G3) *Confirmed.* Features from 1024 unlabelled images give 0.496 at $c = 2$, against 0.505 from 50,000. Claim 1 holds for classification: the unlabelled data a readout needs are fixed by the tied kernel, and 10^3 images suffice. (G5) *Confirmed.* The Bayes grid head reaches 0.41 at $c = 2$, against ≤ 0.25 for every single top variable. It is also the most label-efficient readout: 0.34 with 300 labels and 0.38 with 1000. (G4) *Refuted.* Backpropagation through the same architecture is behind at $P = 300$ (0.283, against 0.290 for the untied head and 0.337 for the Bayes grid head). It leads from $P = 1000$ to 10^4 by up to 0.05 (0.425 against 0.379–0.396 at 3000), and it is behind again with all labels (0.483 against 0.505; with 3000 steps it may be under-trained there). Backprop also had 5000 extra labels for learning-rate choice and early stopping, which the EM heads did not use; the refutation stands without that.

Why (G4) failed: tying shortens the label's path for backpropagation too. Corollary 63(c) argued from the label reach: the signal of the label falls about twentyfold per level below it. That is true per node. But the kernel is tied, so backpropagation needs to reach only the *first* level below the grid. Every level's edges update the same kernel, and the nearest edges carry an undamped label signal. With the grid at $c = 2$ that is one edge away. The decay argument (and Theorem 41) constrains backpropagation for untied deep parameters, or for labels far above the parameters they must teach; it does not apply to a tied kernel read at a grid just above it. Once the labels number in the thousands, backpropagation then gets what EM does not: types chosen for the class rather than for the pixels. The zero-label kernel spends its sixteen types on explaining pixels, and a linear head on them saturates at 0.38–0.40 for $P \leq 3000$.

What this settles. The strong root is the grid, and it can be built without labels: 0.51–0.53 linear and 0.41 generative, from features that 10^3 unlabelled images fix. That is well above every root variable (≤ 0.38), and above a linear head on raw pixels (0.39). Labels are needed only for the head, and the generative head is best when labels are few. Claim 2, in its strong form on CIFAR, does not hold for this architecture. Local EM beats backpropagation only below about 10^3 labels. Between 10^3 and 10^4, backpropagation through the same tied tree uses the labels better, because it makes class-relevant types and tying brings the label within one edge of the kernel. The advantage of the local rule on CIFAR is claim 1: zero labels and 10^3 images for the representation.

10.15 What a label can buy, wherever it sits

The natural pure repair is to make the label the *parent* of the grid in a joint model, GP(c) above a tied tree, and fit everything by EM on labelled and unlabelled images together. The label would then vote on n_c region types instead of one root type. The theory rules this out before it is built.

Proposition 64 (a label is worth at most $\log C$, wherever it sits)**.** *For any joint model $p_\theta(X, y)$, whatever the position of y, let $\hat{\theta}$ maximize the joint log-likelihood on N images of which P are labelled, and let $\hat{\theta}_0$ maximize the pixel log-likelihood. Then $\mathcal{L}_X(\hat{\theta}_0) - \mathcal{L}_X(\hat{\theta}) \leq P \log C$: the joint optimum is within $(P/N) \log C$ nats per image of a pixel-optimal model.*

Proof. $\log p(X, y) = \log p(X) + \log p(y \mid X)$ for every model, so the joint log-likelihood is the pixel log-likelihood plus the conditional log-likelihood of the labels. The argument of Proposition 54 applies unchanged. □

Corollary 65 (when a generative rule can beat a discriminative one)**.** *The classifier of the joint optimum is the best one among models within $(P/N) \log C$ nats per image of the pixel optimum.*

If the class is a function of the pixel-optimal state, it reaches the Bayes rule from few labels, whatever the depth. This is the tied grammar, where the class is the root of the context-optimal state (Theorem 21), and the local rule wins exponentially in depth (§9). If the class is not such a function, labels cannot move the representation toward it by more than $(P/N)\log C$ per image. A rule that ignores the pixels (conditional likelihood only) has no such cap and pays only in label variance.

On CIFAR with $P = 1000$ of $N = 50{,}000$, the cap is 0.046 nats per image. Moving the class-mixture of GP(c) from the pixel-optimal partition (colour and texture, §10.12) to the class costs far more, and so would sharpening the band of types below the grid. So the label-as-parent joint EM is predicted to equal the two-stage readout of §10.14 at small P, and to lose to backpropagation between 10^3 and 10^4 labels. We do not run it.

Proposition 66 (the label gradient is local)**.** *In any tree model with the label as a node, $\partial_\theta \log p(y \mid X)$ is the difference of the grid-local expected statistics with the label clamped and with it free (Corollary 56(b)). Two sum–product passes compute it, each node using only its own posterior and its parent pair.*

So on a tree, "backpropagation from labels" and "local learning" do not differ in locality: conditional maximum likelihood is itself a clamped-minus-free local rule. They differ in what they learn from. The only resource a local rule has beyond the labels is the unlabelled data, and Proposition 64 bounds what a generative objective can extract from it for a class that the pixels do not favour. On such data, claim 2 reduces to two estimators in sequence. First, maximum likelihood on the unlabelled images fixes the representation (claim 1: 10^3 images). Second, conditional maximum likelihood by the clamped-minus-free rule refines it from the labels. The comparison with backpropagation from scratch is then a comparison of starting points on the same objective. (The head-to-head of §10.14 used a discriminative linear head on the messages rather than the generative head of GP(c). Its gradient also travels only along tree edges, but it is not literally the clamped-minus-free difference.)

11 Discussion and what remains open

What the symmetry buys. In the untied model the top level is a grid without context, so labels are necessary (Theorem 2). Tying the rules lets the top level borrow the bottom level's context: the s^{L-1} bottom grids of a single example are s^{L-1} conditionally independent views of the same local problem. This is the formal version of the notes' "the substrate is identical, so a strategy learned at one level transfers to all". The learning rule is not gradient-based. It is the statistical Myhill–Nerode quotient (merge grids no context can distinguish), followed by naming through self-consistency across scales.

Limitations. (i) Theorem 4 assumes a non-degenerate realized histogram; Prop. 6 certifies this only via a conservative worst-case λ. (ii) Brute-force naming costs $v!$. Constraint propagation (as in isomorphism testing of random structures) should be polynomial on typical instances, but we have not proved it. (iii) The RHM is noise-free and unambiguous. Noisy or ambiguous grammars need the soft version (belief propagation; srg.pdf, §5). (iv) Experiments use one small configuration.

Open problem 67 (a lower bound for backprop on the tied model; partly addressed in §9)**.** Show that SGD on a deep CNN with root labels (tied or untied weights) needs a number of labelled examples growing with L on the tied RHM, e.g. via the decay of label–tuple correlations $\leq \lambda^{L-1}$

[1]. Together with Corollary 9, this would be a strict separation between backprop and a symmetry-exploiting local rule.

Open problem 68 (polynomial naming)**.** Prove that naming by self-similarity can be done in polynomial time with high probability over the random grammar.

Open problem 69 (soft and ambiguous grammars)**.** Extend §6 to noisy rules, where exact parsing is replaced by belief propagation and deviations are soft. (The fully tied case is covered by §8: context identifies beliefs, Theorem 21.) Does depth–frequency cancellation (Lemma 12) survive?

Open problem 70 (anchorless classes)**.** Show that cross-level consistency identifies the deviations of classes that keep no base rule at some height, pushing the learner's breakdown in §6 from $\rho \approx 0.5$ towards $\rho \to 0$. Is the floor of Theorem 11 attainable for every $\rho > 0$?

Open problem 71 (estimating ρ on real data)**.** The partial-tying model gives an operational definition of how self-similar a dataset is. Estimate the effective ρ of natural images or text by matching learned synonym partitions across scales (the statistic $S(\pi)$ of Prop. 14).

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